Mathematical background

Branching, quasi-stationarity, catastrophe and absorption

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Chapter 2

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Contents

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Mathematical background

2.1 Introduction

The processes studied in this thesis describe populations small enough that their fate is decided by chance rather than by their mean behaviour — a founding cohort of a few individuals, or the contents of a single compartment that may later rupture. At those counts a deterministic rate equation answers the wrong question. It returns a single trajectory, growing or decaying, when what one wants to know is whether the population establishes itself at all, and how large it is if it does. Both are questions about a distribution, and neither is visible in a mean.

Two threads run through this chapter. They are worth separating at the outset, because almost every later section belongs to one or the other.

The first is *conditioning on survival*. A subcritical branching process dies out with probability one, so its unconditional mean decays geometrically and records little beyond the certainty of extinction. Restrict attention to those realisations that happen still to be alive at generation n, however, and the picture changes entirely: the conditional population settles to a fixed distribution, and the mean of that distribution is a constant. For the binary Galton–Watson process with replication probability $p < \frac{1}{2}$ the constant is $1/A(p)$, where

$$A(p) = \lim_{n\to\infty} \frac{S_n}{(2p)^n}$$

and S_n is the probability of survival to generation n. A large part of this chapter is about $A(p)$: what it is, how to compute it, and why no elementary formula for it has been found despite a determined search.

The second thread is the *near-critical regime*. As p rises towards $\frac{1}{2}$ the process takes ever longer to die, the quasi-stationary mean $1/A(p)$ diverges, and any numerical scheme that works by iterating to stationarity becomes impractical. Just above $\frac{1}{2}$ the size of the founding cohort begins to weigh heavily on whether the process survives at all. The behaviour on either side of $p = \frac{1}{2}$ is what makes small founding populations interesting, and it is also what makes them awkward to compute with.

Section 2.2 assembles the standard apparatus: exponential clocks and the races between them, continuous-time Markov chains and their generators, probability generating functions, and the linear birth–death process, whose generating-function partial differential equation is the template for everything in section 2.7. Section 2.3 sets out the Galton–Watson process and establishes the two facts on which the rest depends — that extinction is certain at and below criticality, and that at criticality the expected lifetime is nevertheless infinite. Section 2.4 derives the limiting conditional mean and variance, and so introduces $A(p)$. Section 2.5 asks what changes when the process starts from a cohort of k individuals rather than one, treats a discrete birth–death process subject to catastrophe, obtains the corresponding constant for the continuous-time birth–death process — where a closed form *is* available — and defines the continuous-time birth–death–catastrophe process used in later chapters. Section 2.6 is devoted to $A(p)$ itself. Section 2.7 then changes tack and develops the method of characteristics for the generating-function equations of absorption models.

Two remarks on how the chapter is written. First, section 2.6 is not background. The infinite product, the exact series and its two-sided bounds, the parity bound, the near-critical asymptotic and the identification of $A(p)$ with a value of the Koenigs linearising function of the logistic map are results of this thesis, placed here rather than later because the quasi-stationary material of section 2.4 reduces to that single constant and because later chapters use it throughout. The same applies to the closed form obtained in section 2.7.4. Everything else in the chapter is standard, and is presented as such. Second, the argument is in places heuristic — continuum approximations to difference equations, mean-field substitutions, asymptotic matching. Every such step is marked where it occurs, and where a claim can be checked numerically it has been.

2.2 Preliminaries

This section fixes notation and collects the standard results used later. Nothing in it is new, and a reader fluent in continuous-time Markov chains and generating functions may go straight to section 2.3. It is included because three things are used repeatedly and are easier to state once: that the competition between exponential clocks converts rates into probabilities, that a system of forward equations for state probabilities becomes a single partial differential equation for a generating function, and that the linear birth–death process is solvable by exactly the route that section 2.7 will push further. Standard treatments are Norris [1], Karlin and Taylor [2], and, closest to the applications here, Allen [3].

2.2.1 Exponential clocks

Every model in this thesis is built from events whose rate of occurrence does not depend on how long the system has been waiting for them. That single assumption forces the exponential distribution.

Definition 2.1 (Exponential law)**.** A random variable T is exponentially distributed with rate $\theta > 0$, written $T \sim \mathrm{Exp}(\theta)$, if it has density and distribution function

$$f_T(t) = \theta \mathrm{e}^{-\theta t}, \qquad F_T(t) = 1 - \mathrm{e}^{-\theta t}, \qquad t \geq 0,$$

and $f_T(t) = 0$ for $t < 0$. Its mean is $1/\theta$ and its variance $1/\theta^2$.

The property that matters is memorylessness. For $t_1, t_2 > 0$,

$$\mathbb{P}(T > t_1 + t_2 \mid T > t_1) = \frac{\mathrm{e}^{-\theta(t_1+t_2)}}{\mathrm{e}^{-\theta t_1}} = \mathrm{e}^{-\theta t_2} = \mathbb{P}(T > t_2)\,, \tag{2.1}$$

so a clock that has already run for time t_1 is statistically indistinguishable from a fresh one. The exponential law is the only continuous distribution with this property, which is why a process assembled from constant-rate events depends on its present state alone and not on its history.

When several events compete, the following two facts organise everything that follows. Let $T_1, \ldots, T_m$ be independent with $T_i \sim \mathrm{Exp}(\theta_i)$, and let $T = \min_i T_i$ be the time of the first event.

Proposition 2.2 (Competing clocks). *With the notation above, write* $\Theta = \sum_{i=1}^{m} \theta_i$. *Then*

$$T \sim \mathrm{Exp}(\Theta), \qquad \mathbb{P}(T_k = T) = \frac{\theta_k}{\Theta}, \tag{2.2}$$

and the identity of the winning clock is independent of T.

Proof. For the first claim, $\mathbb{P}(T > t) = \prod_i \mathbb{P}(T_i > t) = \prod_i \mathrm{e}^{-\theta_i t} = \mathrm{e}^{-\Theta t}$. For the second, condition on the value of T_k and use independence:

$$\mathbb{P}(T_k = T) = \int_0^\infty \theta_k \mathrm{e}^{-\theta_k u} \prod_{i \neq k} \mathrm{e}^{-\theta_i u}\,\mathrm{d}u = \theta_k \int_0^\infty \mathrm{e}^{-\Theta u}\,\mathrm{d}u = \frac{\theta_k}{\Theta}.$$

Repeating the calculation with the integral truncated at t gives $\mathbb{P}(T_k = T,\ T > t) = (\theta_k/\Theta)\mathrm{e}^{-\Theta t} = \mathbb{P}(T_k = T)\,\mathbb{P}(T > t)$, which is the asserted independence. □

Proposition 2.2 is used in two ways. It says that a rate θ_k becomes a probability θ_k/Θ as soon as one asks which of several things happened rather than when; section 2.5.4 uses exactly this to identify the replication probability p of a discrete-generation model with $\lambda/(\lambda+\mu)$ in a continuous-time one. And it is the whole content of the direct simulation method.

Remark 2.3 (Direct simulation). Proposition 2.2 is an algorithm as well as a theorem. Given the current state, compute the rate θ_i of every possible event and their sum Θ; draw $T \sim \mathrm{Exp}(\Theta)$ by inverting the distribution function of definition 2.1 on a uniform variate; draw the event k with probability θ_k/Θ; advance the clock by T, apply event k, and repeat. Because the two draws are independent and the exponential law is memoryless, the resulting trajectory has exactly the law of the process. This is Gillespie's direct method [4]; it is used in this thesis to check analytical results, never to replace them.

2.2.2 Continuous-time Markov chains

Let $\{X_t : t \geq 0\}$ be a stochastic process taking values in a countable state space $\mathcal{S}$, which throughout will be a set of population counts or of tuples of counts. The process is a *continuous-time Markov chain* if, for all times $s, t \geq 0$ and all states,

$$\mathbb{P}(X_{t+s} = j \mid X_t = i,\ \{X_u\}_{u<t}) = \mathbb{P}(X_{t+s} = j \mid X_t = i)\,, \tag{2.3}$$

so that the past influences the future only through the present. The chain is *time homogeneous* if the right-hand side depends on s but not on t, in which case one writes $p_{ij}(s) = \mathbb{P}(X_{t+s} = j \mid X_t = i)$. All chains in this thesis are time homogeneous unless a later chapter says otherwise , and the time-inhomogeneous case is taken up in the later chapter on non-constant rates.

Such a chain is specified by its *generator*, the matrix $Q = (q_{ij})$ whose off-diagonal entries are the rates of the corresponding transitions,

$$q_{ij} = \lim_{\Delta t \to 0} \frac{p_{ij}(\Delta t)}{\Delta t} \quad (j \neq i), \qquad q_{ii} = -\sum_{j \neq i} q_{ij} =: -q_i, \tag{2.4}$$

so that every row of Q sums to zero. Equivalently, and this is the form used to define models below,

$$\mathbb{P}(X_{t+\Delta t} = j \mid X_t = i) = q_{ij}\,\Delta t + o(\Delta t) \qquad (j \neq i), \tag{2.5}$$

with the complementary probability $1 - q_i \Delta t + o(\Delta t)$ of no transition at all. By proposition 2.2 the chain leaves state i after a holding time distributed as $\mathrm{Exp}(q_i)$ and then jumps to j with probability q_{ij}/q_i. A state with $q_i = 0$ is *absorbing*: the chain that reaches it stays there forever. Extinction and rupture will both be modelled as absorbing states, and the whole of section 2.4 concerns the behaviour of a chain conditioned on not yet having been absorbed.

Differentiating the Chapman–Kolmogorov identity $p_{ij}(t + \Delta t) = \sum_k p_{ik}(t) p_{kj}(\Delta t)$ with respect to Δt at $\Delta t = 0$ gives the *forward Kolmogorov equations*,

$$\frac{\mathrm{d}p_{ij}(t)}{\mathrm{d}t} = \sum_k p_{ik}(t)\, q_{kj}, \qquad \text{that is} \qquad \frac{\mathrm{d}P}{\mathrm{d}t} = P\,Q. \tag{2.6}$$

Written out for a fixed initial state and with $p_j(t)$ for the probability of occupying state j at time t, (2.6) becomes the master equation

$$\frac{\mathrm{d}p_j(t)}{\mathrm{d}t} = \sum_{k \neq j} q_{kj}\, p_k(t) - q_j\, p_j(t) : \tag{2.7}$$

probability flows into state j from every state that can reach it and out of j at total rate q_j. Every model in this thesis is specified by writing down (2.5) and then reading off (2.7).

2.2.3 Probability generating functions

For a random variable X taking values in $\{0, 1, 2, \dots\}$ with $p_k = \mathbb{P}(X = k)$, the probability generating function is

$$\phi(z) = \mathbb{E}\Big(z^X\Big) = \sum_{k=0}^{\infty} p_k z^k, \tag{2.8}$$

which converges at least for $|z| \leq 1$ since the p_k sum to one. The series determines the distribution — $p_k = \phi^{(k)}(0)/k!$ — and moments are read off at $z = 1$:

$$\phi(1) = 1, \qquad \phi'(1) = \mathbb{E}(X)\,, \qquad \phi''(1) = \mathbb{E}(X(X-1))\,. \tag{2.9}$$

Two structural properties do most of the work. If X and Y are independent then $\phi_{X+Y} = \phi_X \phi_Y$, so sums become products. And if N is an independent count with probability generating function ψ, the random sum $X_1 + \cdots + X_N$ of independent copies of X has probability generating function $\psi \circ \phi$; composition of generating functions is what makes branching processes tractable, and section 2.3 rests on it.

For a pair of counts the definition extends to

$$G(x, y) = \mathbb{E}\Big(x^X y^Y\Big) = \sum_i \sum_j \mathbb{P}(X = i,\, Y = j)\ x^i y^j, \tag{2.10}$$

with marginals recovered by setting the other variable to one. The absorption models of section 2.7 track particles inside and outside a compartment and therefore need this bivariate form; section 2.B records how to invert it.

When the counts evolve in time, G acquires a time argument, $G(x, y, t) = \mathbb{E}\Big(x^{X_t} y^{Y_t}\Big)$, and the master equation (2.7) for the infinite family of state probabilities collapses to a single partial differential equation for G. The mechanism is uniform: multiply the equation for $p_{ij}(t)$ by $x^i y^j$, sum over all states, shift indices so that every sum is again of the form (2.10), and recognise the factors of i and j produced by the linear rates as derivatives ∂_x and ∂_y. Because the rates in these models are linear in the counts, the resulting equation is always of first order, and the method of characteristics therefore always applies in principle. Section 2.2.4 carries this out in the simplest case; section 2.7 does it three times over.

2.2.4 The linear birth–death process

The linear birth–death process is the continuous-time counterpart of the Galton–Watson process, and it is the last piece of standard machinery needed. Let X_t count individuals, each of which independently gives birth at rate λ and dies at rate μ. In the notation of (2.5) the non-zero rates out of state n are

$$q_{n,n+1} = \lambda n, \qquad q_{n,n-1} = \mu n, \tag{2.11}$$

so that $q_n = (\lambda + \mu)n$ and $n = 0$ is absorbing. The master equation (2.7) reads

$$\frac{\mathrm{d}p_n(t)}{\mathrm{d}t} = \lambda(n-1)p_{n-1}(t) + \mu(n+1)p_{n+1}(t) - (\lambda+\mu)n\, p_n(t). \tag{2.12}$$

Multiplying by z^n and summing over $n \geq 0$, the three terms become $\lambda z^2 \partial_z G$, $\mu\, \partial_z G$ and $-(\lambda+\mu)z\, \partial_z G$ after the index shifts described above, so that

$$\frac{\partial G}{\partial t} = (\lambda z^2 - (\lambda+\mu)z + \mu)\frac{\partial G}{\partial z} = (\lambda z - \mu)(z-1)\frac{\partial G}{\partial z}, \qquad G(z,0) = z^N, \tag{2.13}$$

for a process started from N individuals. The quadratic coefficient factorises with roots $z = 1$ and $z = \mu/\lambda$; the same factorisation, in a different variable, reappears in section 2.7.4 and is what makes that calculation possible. Solving (2.13) along its characteristics gives the classical result of Kendall [5],

$$G(z,t) = \left(\frac{\mu(1-z) - (\mu - \lambda z)\mathrm{e}^{-(\lambda-\mu)t}}{\lambda(1-z) - (\mu-\lambda z)\mathrm{e}^{-(\lambda-\mu)t}}\right)^N, \qquad \lambda \neq \mu. \tag{2.14}$$

Two consequences are needed later. The mean follows from (2.9) or, more quickly, from the fact that each individual contributes independently:

$$\mathbb{E}(X_t) = N\mathrm{e}^{(\lambda-\mu)t}. \tag{2.15}$$

The probability of extinction by time t is $p_0(t) = G(0,t)$, that is

$$p_0(t) = \begin{cases} \left(\dfrac{\mu - \mu\mathrm{e}^{(\mu-\lambda)t}}{\lambda - \mu\mathrm{e}^{(\mu-\lambda)t}}\right)^N, & \lambda \neq \mu, \\[2ex] \left(\dfrac{\lambda t}{1+\lambda t}\right)^N, & \lambda = \mu, \end{cases} \tag{2.16}$$

the critical case following by taking $\mu \to \lambda$ in the first line. Letting $t \to \infty$ recovers the familiar threshold: extinction is certain when $\lambda \leq \mu$, and occurs with probability $(\mu/\lambda)^N$ when $\lambda > \mu$. Section 2.5.4 takes (2.16) as its starting point.

2.3 The Galton–Watson process

The Galton–Watson process is a discrete-time Markov chain on the non-negative integers in which every member of a population independently either leaves offspring or does not, at each generational increment. It is the oldest and the simplest model of stochastic branching, and it remains the natural place to begin, because the question it was invented to answer — whether a line of descent survives — is the question this thesis asks throughout. The standard references are Harris [6] and Athreya and Ney [7].

Suppose each individual produces, independently of every other and at the end of its life, a number of offspring distributed as the random variable L, with

$$\mathbb{P}(L = k) = p_k, \qquad k = 0, 1, 2, \ldots,$$

and write $\mu = \mathbb{E}(L)$ for the mean number of offspring per individual. Let Z_n denote the size of the nth generation, started from a single progenitor, $Z_0 = 1$, so that

$$Z_{n+1} = \sum_{i=1}^{Z_n} L_i, \tag{2.17}$$

the L_i being independent copies of L. Conditioning on the previous generation and using the tower property,

$$\mathbb{E}(Z_{n+1}) = \mathbb{E}(\mathbb{E}(Z_{n+1} \mid Z_n)) = \mathbb{E}(\mu Z_n) = \mu \mathbb{E}(Z_n)\,,$$

so $\mathbb{E}(Z_n) = \mu^n$. When $\mu < 1$ the expected population decays geometrically, and since Z_n is integer-valued, Markov's inequality gives $\mathbb{P}(Z_n \geq 1) \leq \mu^n \to 0$: extinction is certain.

Generating functions convert (2.17) into an iteration. Write $\phi(z) = \mathbb{E}\left(z^L\right)$ for the probability generating function of the offspring law. Because Z_{n+1} is a random sum of independent copies of L, the composition rule of section 2.2.3 gives $\phi_{n+1} = \phi_n \circ \phi$, where ϕ_n is the probability generating function of Z_n; that is, ϕ_n is the n-fold composition of ϕ with itself. Setting $z = 0$ and writing $u_n = \mathbb{P}(Z_n = 0)$ for the probability of extinction by generation n, the same composition yields the iteration on which almost everything below depends,

$$u_{n+1} = \phi(u_n), \qquad u_0 = 0. \tag{2.18}$$

2.3.1 Death or division

Throughout this thesis we specialise to the binary, or *simple*, case, in which an individual either divides into two or dies:

$$\begin{aligned} p_2 &= p && \text{(division)},\\ p_0 &= 1 - p && \text{(death)},\\ p_i &= 0 && \text{for all } i \neq 0, 2. \end{aligned}$$

The mean number of offspring is $\mu = 2p$, and the expected population descended from a single ancestor is

$$\mathbb{E}(Z_n) = (2p)^n. \tag{2.19}$$

The process is *subcritical* when $2p < 1$, *critical* when $p = \frac{1}{2}$ and *supercritical* when $2p > 1$. Figure 2.1 shows realisations either side of the critical point.

The offspring probability generating function is

$$\phi(z) = \mathbb{E}\left(z^L\right) = pz^2 + (1 - p), \tag{2.20}$$

so that (2.18) reads

$$u_{n+1} = p\,u_n^2 + 1 - p. \tag{2.21}$$

The recursion has a direct combinatorial reading, and it is worth taking a moment over it because the same argument recurs in sections 2.5.1 and 2.5.3. If the founder dies at its first step, which happens

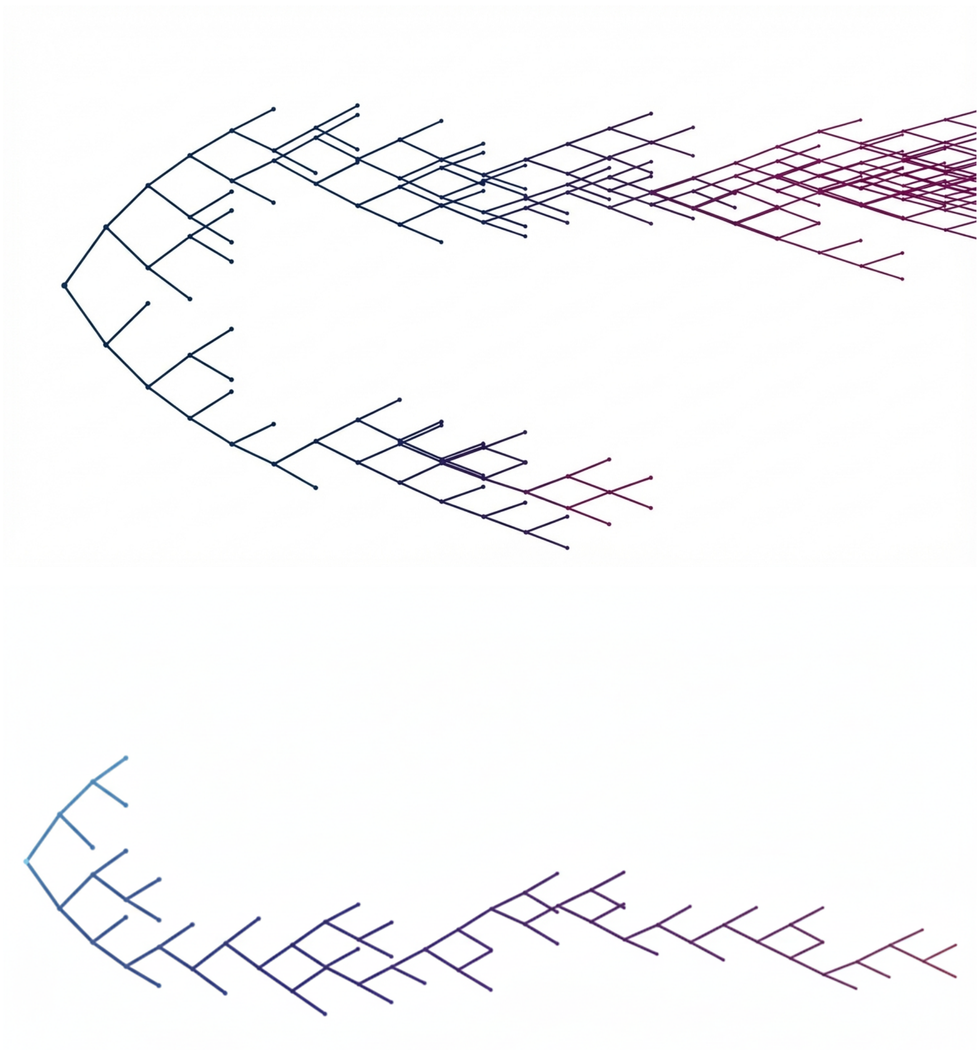


Figure 2.1: Realisations of the simple Galton–Watson process, drawn as genealogies with generation increasing to the right. Top: supercritical, $p = 0.58$, where the number of lines grows without bound. Bottom: subcritical, $p = 0.47$; the same construction, but the tree terminates. Colour encodes generation number.

with probability $1 - p$, extinction has already occurred. If instead it divides, with probability p, then extinction by generation $n + 1$ requires each of the two fresh lineages to die out within its own n generations, and those lineages are independent, so that happens with probability u_n^2.

It is more convenient to work with the complement. Put

$$S_n = 1 - u_n = \mathbb{P}(Z_n > 0)\,,$$

the probability that the process is still alive at generation n. Substituting $u_n = 1 - S_n$ into (2.21) and simplifying gives

$$S_{n+1} = 2pS_n - pS_n^2, \qquad S_0 = 1. \tag{2.22}$$

A population cannot return from extinction, so u_n is non-decreasing and bounded above by 1; equivalently S_n is non-increasing and bounded below by 0, and therefore converges to some $S_\infty \in [0,1]$.

2.3.2 Certainty of extinction

Since S_n converges, its limit must be a fixed point of (2.22). Setting $S_{n+1} = S_n = S_\infty$,

$$S_\infty = 2pS_\infty - pS_\infty^2 \qquad \Longleftrightarrow \qquad p\,S_\infty\left(S_\infty + \tfrac{1}{p} - 2\right) = 0,$$

with roots

$$S_\infty = 0 \qquad \text{and} \qquad S_\infty = 2 - \frac{1}{p} = \frac{2p-1}{p}.$$

For $p < \frac{1}{2}$ the second root is negative and carries no probabilistic meaning, so the only admissible limit is $S_\infty = 0$ and extinction is certain. At $p = \frac{1}{2}$ the two roots collide at the origin and extinction remains certain. For $p > \frac{1}{2}$ a positive root appears, and it is that root which is attained: the process survives forever with positive probability $S_\infty = (2p-1)/p$. Figure 2.2 shows the transition, together with the change in stability of the origin that produces it.

2.3.3 The critical case: a power law and an infinite expected lifetime

At $p = \frac{1}{2}$ the expectation is constant, $\mathbb{E}(Z_n) = 1$ for every n, and yet extinction is certain. The two facts are reconciled by the tail: rare realisations survive for enormously long times and grow enormously large, and they do so at exactly the rate needed to hold the mean at one. Demonstrating this is worth the space, because the same calculation supplies the asymptotics of S_n that section 2.6 will need.

Let

$$T = \inf\{n : Z_n = 0\}$$

be the extinction time, so that the process passes through generations $0, 1, \ldots, T-1$ and is dead at T. Writing T as a sum of indicators and taking expectations term by term — legitimate because every term is non-negative —

$$\mathbb{E}(T) = \sum_{n=0}^{\infty} \mathbb{P}(T > n) = \sum_{n=0}^{\infty} \mathbb{P}(Z_n > 0) = \sum_{n=0}^{\infty} S_n. \tag{2.23}$$

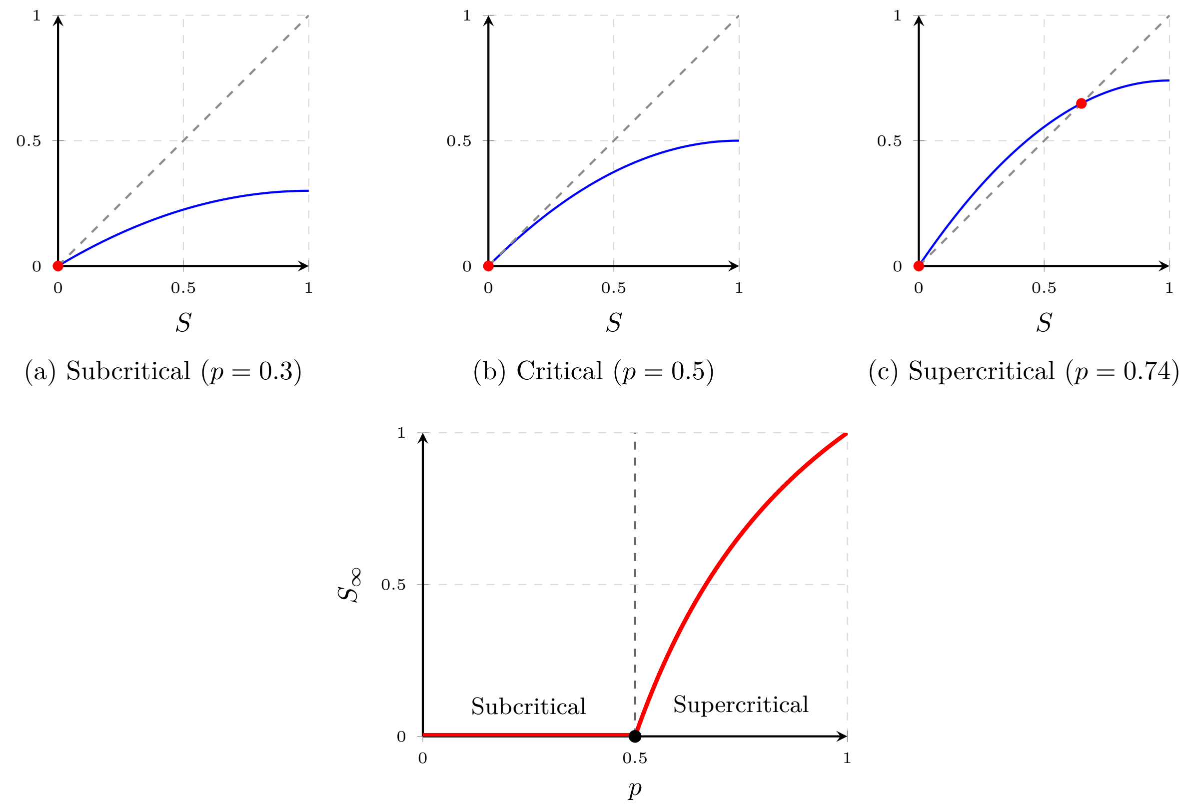
The question is therefore how fast S_n decays at criticality.

Setting $p = \frac{1}{2}$ in (2.22) gives $S_{n+1} - S_n = -\frac{1}{2}S_n^2$. Once n is large the unit generational increment is small compared with the scale on which S_n varies, and the difference on the left is well approximated by a derivative. This step is heuristic — it replaces a difference equation by a differential one, and nothing here justifies the replacement beyond the smallness of the increment relative to the solution — but its conclusion is standard and is confirmed by simulation below. With that caveat,

$$\frac{\mathrm{d}S}{\mathrm{d}n} = -\tfrac{1}{2}S^2, \qquad \text{with general solution} \qquad S(n) = \frac{2}{n + \text{const}},$$

so that

$$S_n \sim \frac{2}{n} \qquad (n \to \infty), \tag{2.24}$$



(a) Subcritical ($p = 0.3$) (b) Critical ($p = 0.5$) (c) Supercritical ($p = 0.74$)

(d) The transition in S_∞

Figure 2.2: Fixed points of the survival recursion (2.22). Panels (a)–(c) show the map $S \mapsto 2pS - pS^2$ (blue) against the diagonal (dashed), with the fixed points marked in red. Below and at criticality the origin is the only fixed point in $[0, 1]$ and it is attracting, so extinction is certain; at $p = \frac{1}{2}$ the map is tangent to the diagonal there. For $p > \frac{1}{2}$ the origin turns repelling and a stable positive fixed point $S_\infty = (2p - 1)/p$ separates from it, traced in panel (d). The survival probability is identically zero below the critical point and rises continuously above it.

as figure 2.4(a) confirms. Substituting into (2.23), the tail of the sum behaves like twice the harmonic series,

$$\mathbb{E}(T) = \sum_{n=0}^{\infty} S_n \sim \sum_{n=1}^{\infty} \frac{2}{n},$$

which diverges (figure 2.3). Since every S_n is positive, $\mathbb{E}(T) = \infty$: the expected lifetime of the critical Galton–Watson process is infinite even though extinction occurs with probability one.

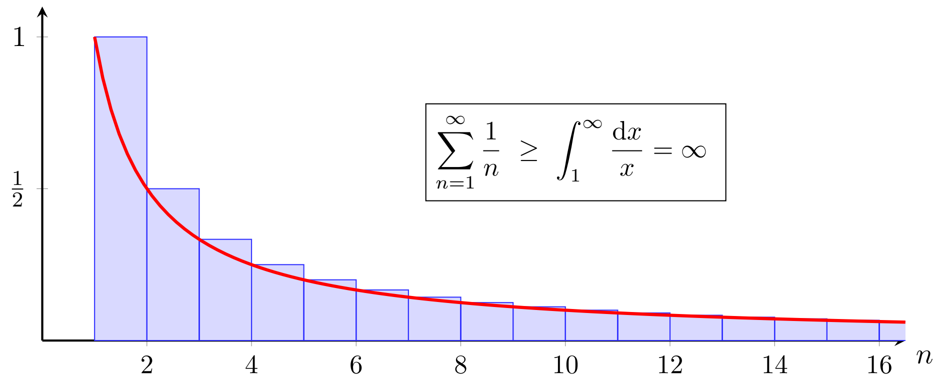


Figure 2.3: Divergence of the harmonic series. Each blue rectangle has area $1/n$ and contains the region under the red curve $1/x$ on $[n, n+1]$, so the sum of the areas exceeds $\int_1^{\infty} x^{-1}\mathrm{d}x = \infty$. Because $S_n \sim 2/n$ at criticality, the same divergence carries over to $\mathbb{E}(T) = \sum_n S_n$.

The heavy tail responsible is visible directly in simulation. A power-law distributed quantity carries a modest but non-negligible chance of an extraordinarily large realisation, and a sample mean computed from such a quantity does not settle: the larger the sample, the larger the mean, because the occasional enormous value keeps pushing the normalised sum upward. Figure 2.4 collects the three classical critical exponents — for the survival probability, for the extinction time and for the total progeny.

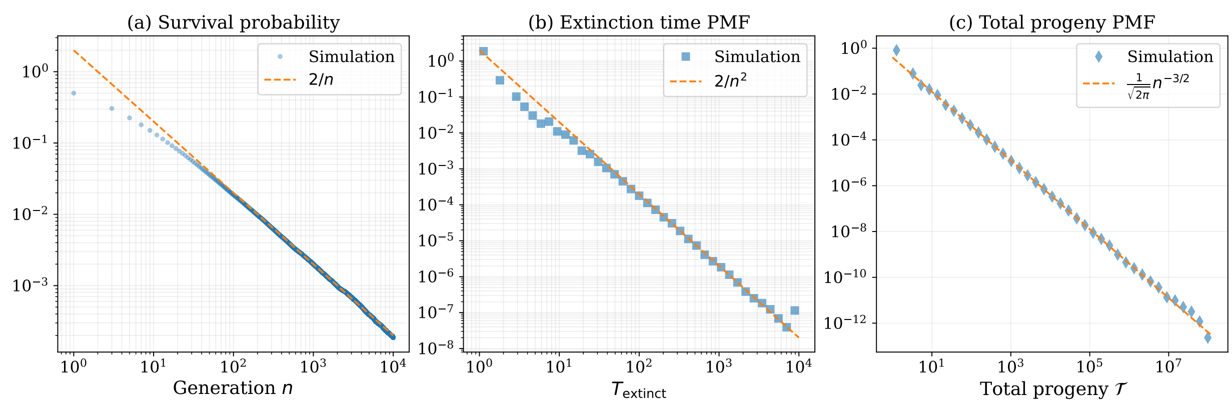


Figure 2.4: Power-law behaviour in the critical Galton–Watson process ($p = 0.5$), from 10^6 realisations. (a) Survival probability $S_n = \mathbb{P}(Z_n > 0)$, showing the late-generation scaling $S_n \sim 2/n$ of (2.24). (b) Probability mass function of the extinction time $T = \inf\{n : Z_n = 0\}$, with heavy tail $\mathbb{P}(T = n) \sim 2/n^2$. (c) Probability mass function of the total progeny $\mathcal{T} = \sum_{k \geq 0} Z_k$, displaying the classical critical branching asymptotic $\mathbb{P}(\mathcal{T} = n) \sim n^{-3/2}$.

2.4 The limiting conditional distribution

Absorption is often inevitable, as it is here for $p < \frac{1}{2}$, and yet a process on its way to absorption need not be featureless. In many cases it settles, long before it is absorbed, into a distribution that no longer changes with time — a distribution describing the population conditional on its not yet having been absorbed. The phenomenon is *quasi-stationarity*, and the distribution is the quasi-stationary, or Yaglom, limit [8]; modern accounts are given by Méléard and Villemonais [9] and by Collet, Martínez and San Martín [10]. This section establishes it for the binary Galton–Watson process by computing the limiting conditional mean and variance, and in doing so introduces the constant that section 2.6 is devoted to.

2.4.1 Late-time survival probability

Everything follows from the behaviour of S_n. Take $p < \frac{1}{2}$, so that death of an individual is more probable than its division. Then $S_n \to 0$, and as it does so the quadratic term in

$$S_{n+1} = 2pS_n - pS_n^2$$

becomes negligible beside the linear one. The recursion is therefore increasingly well approximated by

$$S_{n+1} \approx 2p\, S_n, \tag{2.25}$$

and so at late times

$$S_n \sim A(p)\,(2p)^n \qquad (n \to \infty), \qquad A(p) = \lim_{n\to\infty} \frac{S_n}{(2p)^n}. \tag{2.26}$$

The constant $A(p)$ absorbs the cumulative effect of the quadratic term over the early generations, before the linear regime takes hold. It is precisely what remains of the conditions imposed on S_n before the recursion attenuates to (2.25), which is why no argument confined to late times will produce it. That the limit exists and is finite and strictly positive for $p \in [0, \frac{1}{2})$ is proved in section 2.6.1 by way of an infinite product, and section 2.6.2 locates it in $(0, \frac{1}{2}]$.

Remark 2.4. The relation (2.26) is an asymptotic equivalence, not an identity. The limit of S_n itself is 0 for $p < \frac{1}{2}$, and one must not write $\lim_n S_n = A(p)(2p)^n$, whose right-hand side still depends on n. What is asserted is that the ratio $S_n/(2p)^n$ converges.

2.4.2 Mean

Decompose the expected population size into the contributions from surviving and from extinct realisations:

$$\mathbb{E}(Z_n) = \mathbb{P}(Z_n = 0)\,\mathbb{E}(Z_n \mid Z_n = 0) + \mathbb{P}(Z_n > 0)\,\mathbb{E}(Z_n \mid Z_n > 0)\,.$$

The first term vanishes identically, since $Z_n = 0$ on that event, so with $\mathbb{P}(Z_n > 0) = S_n$ and $\mathbb{E}(Z_n) = (2p)^n$ from (2.19),

$$\mathbb{E}(Z_n \mid Z_n > 0) = \frac{\mathbb{E}(Z_n)}{S_n} = \frac{(2p)^n}{S_n}. \tag{2.27}$$

The numerator is known exactly; all the content is in S_n, and (2.26) supplies it:

$$\mathbb{E}(Z_n \mid Z_n > 0) \xrightarrow[n\to\infty]{} \frac{(2p)^n}{A(p)(2p)^n} = \frac{1}{A(p)}. \tag{2.28}$$

The geometric factors cancel exactly. At late times the surviving population has a constant expected size, and that size is the reciprocal of $A(p)$; accordingly $1/A(p)$ is called the *quasi-stationary mean* below.

Figure 2.5 shows the convergence, and shows at the same time the difficulty that section 2.6 has to confront: the closer p lies to $\frac{1}{2}$, the higher the plateau and the longer the process takes to reach it.

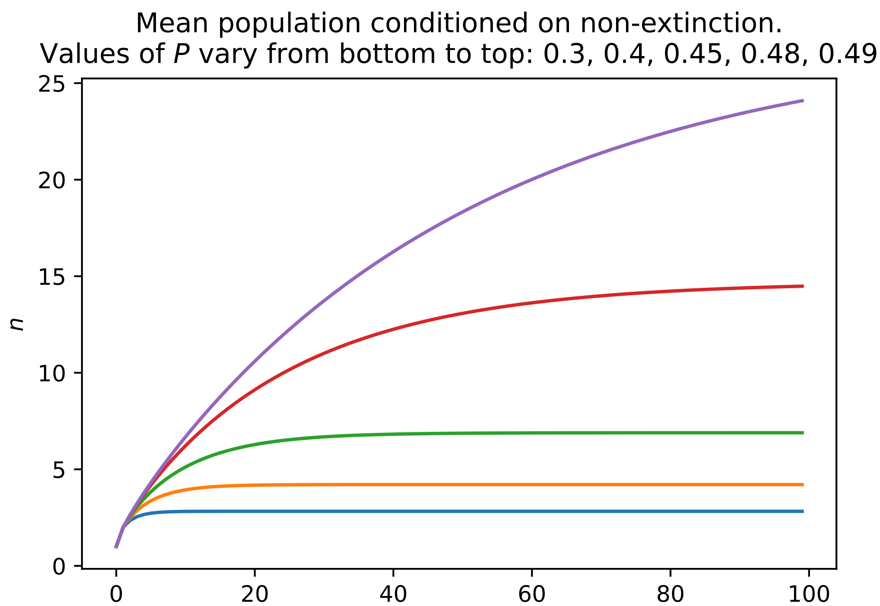


Figure 2.5: The conditional mean $(2p)^n/S_n$ against generation number n, with S_n obtained by iterating (2.22). From bottom to top, $p = 0.3$, 0.4, 0.45, 0.48, 0.49. Each curve approaches the constant $1/A(p)$; the limiting values are 2.825, 4.208, 6.893, 14.703 and 27.476, computed as in section 2.6.7. The lowest three have essentially converged by $n = 100$, but the $p = 0.49$ curve has reached only 24.1 of its limiting 27.5 — an early sign of the near-critical slowdown analysed in section 2.6.7. The vertical axis is the conditional mean.

2.4.3 Variance

The mean alone does not establish quasi-stationarity. To show that the conditional *distribution* settles we need at least the second moment, and computing it is a worthwhile exercise in its own right.

2.4.3.1 The second moment

For a random variable X the moment generating function is $\mathbb{E}\left(\mathrm{e}^{tX}\right)$, which in terms of the probability generating function is $\phi(\mathrm{e}^t)$. Write $M_n(t) = \phi_n(\mathrm{e}^t)$ for the moment generating function of Z_n, so that moments are recovered by

$$\mathbb{E}(Z_n^\rho) = \frac{\mathrm{d}^\rho}{\mathrm{d}t^\rho} M_n(t)\Big|_{t=0}. \tag{2.29}$$

For the binary offspring law (2.20) the first-generation function is

$$M_1(t) = \phi(\mathrm{e}^t) = p\,\mathrm{e}^{2t} + (1-p), \tag{2.30}$$

whose second derivative at $t = 0$ is $4p$, so $\mathbb{E}\left(Z_1^2\right) = 4p$. Reaching generation n requires a little more. Since ϕ_n is the n-fold composition of ϕ,

$$M_n(t) = \phi(\phi(\dots\phi(\mathrm{e}^t)\dots)),$$

and differentiating by the chain rule gives a product of $n+1$ factors,

$$M_n'(t) = \phi'(\phi_{n-1}(\mathrm{e}^t)) \cdot \phi'(\phi_{n-2}(\mathrm{e}^t)) \cdots \phi'(\phi_1(\mathrm{e}^t)) \cdot \phi'(\mathrm{e}^t) \cdot \mathrm{e}^t. \tag{2.31}$$

Now $\phi'(z) = 2pz$ for the binary law, so each factor collapses,

$$\phi'(\phi_k(\mathrm{e}^t)) = 2p\,\phi_k(\mathrm{e}^t) = 2p\,M_k(t), \qquad \phi'(\mathrm{e}^t) = 2p\,\mathrm{e}^t,$$

and (2.31) becomes

$$M_n'(t) = (2p)^n\, M_{n-1}(t)\, M_{n-2}(t) \cdots M_1(t)\, \mathrm{e}^{2t}. \tag{2.32}$$

Before differentiating again, note that every factor on the right equals one at $t=0$: $M_k(0)=1$ for all k, and $\mathrm{e}^0 = 1$. That is what makes the second derivative tractable. Differentiating the product gives a sum of n terms, each identical to (2.32) except that a single factor has been differentiated, and at $t=0$ all the undifferentiated factors become one, leaving

$$M_n''(0) = (2p)^n \left[M_{n-1}'(0) + M_{n-2}'(0) + \cdots + M_1'(0) + 2\right].$$

But $M_k'(0)$ is the mean population at generation k, namely $(2p)^k$, so

$$M_n''(0) = (2p)^n \left[(2p)^{n-1} + \cdots + (2p) + 2\right] = (2p)^n \left[\sum_{i=0}^{n-1} (2p)^i + 1\right].$$

Summing the finite geometric series and rearranging,

$$\mathbb{E}\Big(Z_n^2\Big) = (2p)^n \left(1 + \frac{1}{1-2p}\right) - (2p)^{2n} \left(\frac{1}{1-2p}\right). \tag{2.33}$$

The minus sign is not optional: at $n=1$ the expression must return $\mathbb{E}(Z_1^2) = 4p$, and it does.

2.4.3.2 Late-time variance

From $\mathrm{Var}(Z_n) = \mathbb{E}(Z_n^2) - \mathbb{E}(Z_n)^2$ and (2.33),

$$\mathrm{Var}(Z_n) = (2p)^n \left(1 + \frac{1}{1-2p}\right) - (2p)^{2n} \left(\frac{1}{1-2p}\right) - (2p)^{2n},$$

and for $2p < 1$ the terms in $(2p)^{2n}$ are negligible beside $(2p)^n$ at large n, so $\mathrm{Var}(Z_n) \to (2p)^n(1 + 1/(1-2p))$. It is tempting to read stationarity of the conditional variance straight off that, but the step is not available: the conditional variance is not the unconditional one divided by anything simple. Start instead from the conditional moments, using (2.26) in each:

$$\mathbb{E}\Big(Z_n^2 \mid Z_n > 0\Big) = \frac{\mathbb{E}(Z_n^2)}{S_n} \xrightarrow{n\to\infty} \frac{(2p)^n \left(1 + \frac{1}{1-2p}\right)}{A(p)(2p)^n} = \frac{1}{A(p)} \left(1 + \frac{1}{1-2p}\right),$$

$$\mathbb{E}(Z_n \mid Z_n > 0) = \frac{\mathbb{E}(Z_n)}{S_n} \xrightarrow{n\to\infty} \frac{1}{A(p)}.$$

Both limits are finite, so the conditional variance tends to a constant:

$$\mathrm{Var}(Z_n \mid Z_n > 0) \;\longrightarrow\; \frac{1}{A(p)} \left(1 + \frac{1}{1-2p}\right) - \frac{1}{A(p)^2}. \tag{2.34}$$

The first two moments of the conditional population therefore both stabilise, and the same argument carried to higher moments gives the full quasi-stationary distribution. Note that (2.34), like (2.28), is expressed entirely in terms of p and the single unknown constant $A(p)$. Everything now turns on that constant.

2.4.4 A continuum view of the survival recursion

Before turning to $A(p)$ proper it is useful to record a continuum approximation that recurs twice below. Writing

$$\varepsilon = 1 - 2p \tag{2.35}$$

for the distance from criticality — a symbol used in this sense throughout the chapter — the recursion (2.22) rearranges to

$$S_{n+1} - S_n = -\varepsilon S_n - pS_n^2. \tag{2.36}$$

For large n the unit increment is small on the scale over which S varies, and the left-hand side may be replaced by a derivative. As in section 2.3.3 this replacement is heuristic; what it produces is the Riccati equation

$$\frac{\mathrm{d}S}{\mathrm{d}n} = -\varepsilon S - pS^2. \tag{2.37}$$

The substitution $u = 1/S$ linearises it to $u' = \varepsilon u + p$, with solution $u(n) = K\mathrm{e}^{\varepsilon n} - p/\varepsilon$ and hence

$$S(n) = \frac{\varepsilon}{K\varepsilon\, \mathrm{e}^{\varepsilon n} - p}, \tag{2.38}$$

where K is fixed by the initial condition. Setting $\varepsilon = 0$ recovers the critical scaling $S \sim 2/n$ of (2.24); keeping $\varepsilon > 0$ recovers the geometric decay $S \sim A(p)(2p)^n$ of (2.26), since $(2p)^n = \mathrm{e}^{n \ln(1-\varepsilon)}$ agrees with $\mathrm{e}^{-\varepsilon n}$ to leading order in ε. The constant K is exactly what the continuum approximation cannot supply, because it is determined by the early generations, where the approximation is invalid. That is another way of saying that K is $A(p)$, and that no argument of this kind will produce it. Section 2.6.3 shows how much can nevertheless be extracted near criticality.

2.5 Survival of small populations

Return to the binary process of section 2.3.1 and suppose it is near critical, either just below or just above. With the probability of division p and of death $1 - p$ each close to a half, a lineage dies at its first round almost half the time. Should it survive one generation, both its offspring fail at their own first attempt with probability around 0.25, so the lineage reaches the second generation only with probability about 0.375:

$$0.375 \approx \underbrace{(1 - 0.5)}_{\text{no extinction at the first step}} \times \underbrace{(1 - 0.5 \times 0.5)}_{\text{nor at the second}}.$$

More generally, survival to a given generation is obtained by iterating the probability generating function. With $\phi(z) = pz^2 + (1 - p)$ from (2.20), the extinction and survival probabilities follow from

$$u_{n+1} = p\, u_n^2 + 1 - p, \qquad S_{n+1} = 2pS_n - pS_n^2, \tag{2.39}$$

started at $u_0 = 0$, $S_0 = 1$. At exactly critical branching this gives, for $n = 1, 2, 3, 4$,

$$u_n = 0.500,\ 0.625,\ 0.695,\ 0.741 \qquad (3\ \text{s.f.}),$$

so the probability of survival decays quickly over the first few rounds of division, and does so almost as fast just above criticality as just below. Figure 2.6 makes the point: whatever advantage supercriticality confers, it is not an early-generation advantage.

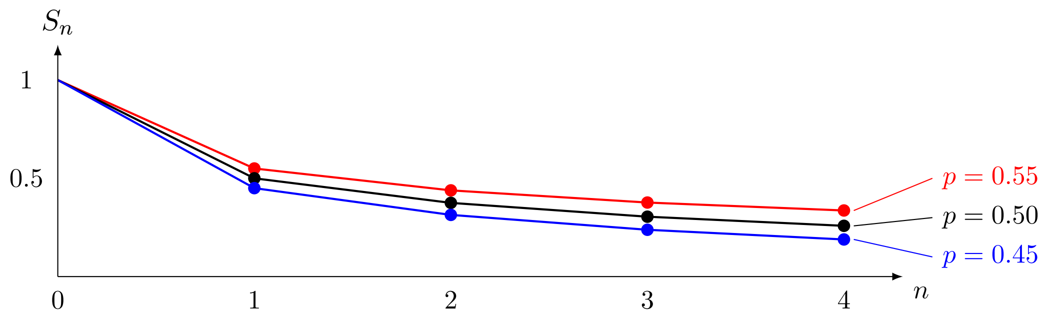


Figure 2.6: Survival probability over the first four generations from a single progenitor, computed from (2.39). Even in the supercritical case ($p = 0.55$) the early decay is steep, and the three curves have barely separated by $n = 4$.

2.5.1 An initial cohort of size k

Now suppose $Z_0 = k$: the process begins not with a single individual but with a group, small to moderate in number, perhaps $k = 10$. Under supercritical branching, growth becomes more predictably exponential as the count rises, because a large number of individuals lets the expected dynamics show through the noise of individual events. The founding cohort is precisely the regime in which that noise dominates instead.

What, then, are such a cohort's chances of long-run persistence? The question is answered by asking for the probability that at least one descendant remains, that is, that not every one of the k founding lineages has died out. The lineages are independent and each is extinct by generation n with probability u_n, so all k have gone with probability u_n^k, and

$$S_n^{(k)} = 1 - u_n^k. \tag{2.40}$$

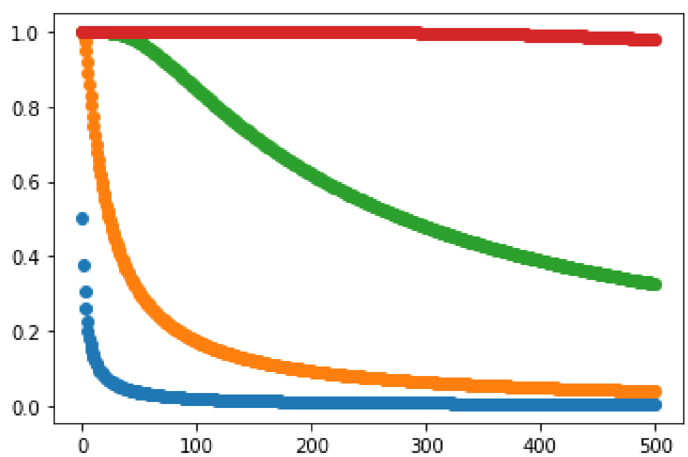


Figure 2.7: Survival probability $S_n^{(k)}$ from (2.40) out to 500 generations at exact criticality, $p = 0.5$, for $k = 1, 10, 100, 1000$ (blue through to red). A larger founding cohort buys longevity, but at criticality every curve is still headed for zero.

Figures 2.7 and 2.8 show $S_n^{(k)}$ through 500 generations for $k = 1, 10, 100, 1000$. The effect visible in figure 2.7 — that a larger first cohort gives improved longevity — is highly sensitive to criticality.

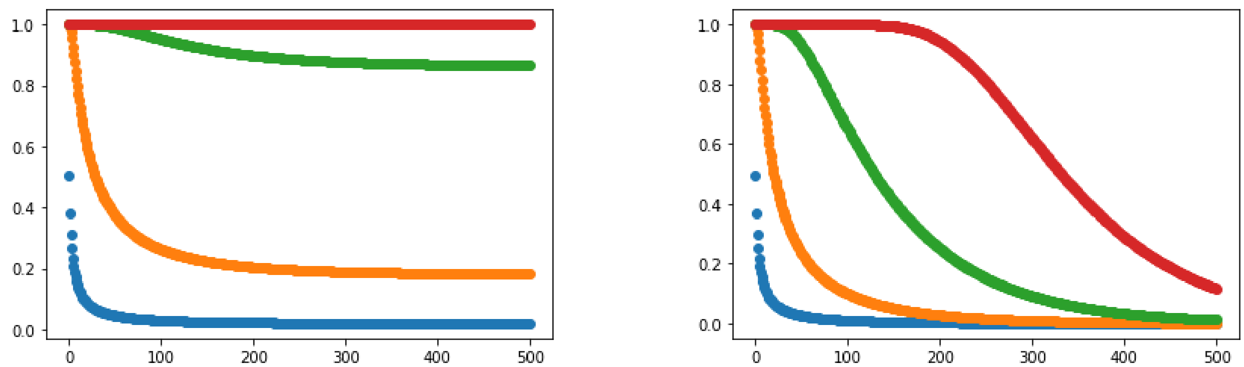


Figure 2.8: As figure 2.7, but with p displaced slightly from criticality: $p = 0.505$ on the left (supercritical), $p = 0.495$ on the right (subcritical). The benefit of a large initial cohort is acutely sensitive to which side of $\frac{1}{2}$ the process sits on. A displacement of half a per cent separates curves that were nearly coincident in figure 2.7.

Persistence improves quickly as p approaches $\frac{1}{2}$ from below and continues to improve dramatically into supercriticality, and the two panels of figure 2.8 differ only in the third decimal place of p.

2.5.2 Conditioning and apparent early growth

The conditioning of section 2.4 has a mirror image which is worth recording, because it shows that the effect is not confined to the subcritical case.

Consider an ensemble of independent homogeneous birth–death processes, all with the same constant rates, and retain only those that happen still to be alive at some large time. The retained subpopulation is automatically enriched for early rapid growth: a realisation that grew quickly at the outset is far more likely to have escaped the early extinction risk documented in figure 2.6, and so far more likely to be in the retained set. No change in the underlying rates is required to produce the enrichment, and the apparent slowdown that follows is not a slowdown at all but a reversion to the mean. The effect is known as the *push of the past*, and was identified in a setting far from the present one, as an explanation for a systematic pattern in longevity and diversity data [11].

That is exactly the conditioning of section 2.4 seen from the other side. There a subcritical process was conditioned on survival and its mean settled to $1/A(p)$ rather than decaying; here a critical or supercritical process is conditioned on survival to a late time and its early growth appears inflated. In both cases the conditioning, not the mechanism, does the work.

The parallel suggests a modelling hypothesis that later chapters take up: that early replication weighs heavily on eventual survival, even when later per-capita rates are unchanged. Figure 2.6 is the reason to take it seriously, since the first few generations are where nearly all of the extinction risk sits.

2.5.3 Discrete birth–death–catastrophe

A birth–death process subject to catastrophe carries, alongside the possibility of extinction by death, the possibility of an abrupt event that removes the population at a stroke. In discrete time and up to the point of catastrophe such a process behaves exactly as the basic Galton–Watson model as far as its limiting conditional distribution is concerned. Conditioning on non-catastrophe, and conditioning also on non-extinction by death, it exhibits a stationary limiting distribution. Whether a quasi-stationary distribution survives when one conditions on *neither* death nor catastrophe is a separate question, and one best settled by simulation. Either way it is useful to have the probability of non-catastrophe in hand.

Probability of catastrophe at step n

Let κ be the probability that a given individual initiates catastrophe at a step. A single individual fails to do so with probability $1-\kappa$, two with $(1-\kappa)^2$, and a cohort of size N with $(1-\kappa)^N$.

Consider first birth and catastrophe alone, with no death. Then the census of the living population at any generation is known with certainty: from a single progenitor the population simply doubles at each step prior to catastrophe. The process therefore comes to its abrupt halt at generation n with probability

$$1-(1-\kappa)^{2^{n-1}}, \tag{2.41}$$

one minus the probability that none of the 2^{n-1} individuals present caused it.

Probability of process survival

Still with no death, define S_n as the probability that the process survives to generation n. Survival to generation 0 is certain, $S_0 = 1$. Survival to generation 1 requires only that the founder divided without catastrophe, $S_1 = 1-\kappa$. Persistence to step 2 requires that the founder divided and that each of its two offspring did too, so $S_2 = (1-\kappa)(1-\kappa)^2$. The pattern is clear: survival to generation n requires every prior step to have been passed, and therefore

$$S_n = \prod_{i=0}^{n-1}(1-\kappa)^{2^i} = (1-\kappa)^{\sum_{i=0}^{n-1} 2^i} = (1-\kappa)^{2^n-1}. \tag{2.42}$$

Death included

The difficulty comes once death is allowed, because the size of the living population is then no longer known with certainty from one generation to the next. The natural guess is that the probabilities shake out when the expected population is used in place of the actual one:

$$S_n \stackrel{?}{=} (1-\kappa)^{\sum_{i=0}^{n-1}\mathbb{E}(Z_i)}. \tag{2.43}$$

This is a heuristic substitution and almost certainly not an identity; it is worth being explicit about why, because the direction of the error can be pinned down exactly. The map $N \mapsto (1-\kappa)^N$ is convex, so Jensen's inequality gives $\mathbb{E}\Big((1-\kappa)^{Z_i}\Big) \geq (1-\kappa)^{\mathbb{E}(Z_i)}$: replacing the random census by its mean systematically understates the chance of getting through a step. Nor are the steps independent, since surviving one is evidence of a smaller population, which raises the chance of surviving the next. Both effects push the same way, so (2.43) is better read as a lower bound than as an equality. How tight a bound remains a question for numerical work.

2.5.4 Continuous-time birth–death

The discrete process resists a closed form for $A(p)$, as section 2.6 shows at length. Its continuous-time counterpart does not, and the comparison between the two turns out to be informative, so the continuous-time constant is recorded here.

Take the linear birth–death process of section 2.2.4, with per-capita birth rate λ and death rate μ, started from N individuals. Its extinction probability $p_0(t)$ is (2.16), and its survival probability is $S^{(N)}(t) = 1 - p_0(t)$.

Limiting mean conditioned on non-extinction

The process has a stationary conditional mean when $\mu > \lambda$, that is in the subcritical case. Exactly as in section 2.4.2,

$$\mathbb{E}(X_t \mid X_t > 0) = \frac{\mathbb{E}(X_t)}{S^{(N)}(t)}, \tag{2.44}$$

and the late-time limit follows on applying L'Hôpital's rule. It is quicker to take $N = 1$ and compute directly. With $\mathbb{E}(X_t) = \mathrm{e}^{(\lambda-\mu)t}$ from (2.15) and

$$S(t) = 1 - \frac{\mu - \mu\mathrm{e}^{(\mu-\lambda)t}}{\lambda - \mu\mathrm{e}^{(\mu-\lambda)t}} = \frac{\lambda - \mu}{\lambda - \mu\mathrm{e}^{(\mu-\lambda)t}},$$

(2.44) gives

$$\mathbb{E}(X_t \mid X_t > 0) = \frac{\lambda\mathrm{e}^{(\lambda-\mu)t} - \mu}{\lambda - \mu}.$$

For $\mu > \lambda$ the exponential decays, and the limiting conditional mean is the constant

$$\lim_{t\to\infty} \mathbb{E}(X_t \mid X_t > 0) = \frac{\mu}{\mu - \lambda} = \frac{1}{1 - \lambda/\mu} \left(= \frac{1}{A_\mathrm{c}}\right). \tag{2.45}$$

To compare this with the discrete-time constant the two parametrisations must be matched, and proposition 2.2 is what matches them: with birth and death clocks running at rates λ and μ, the probability that the birth clock wins is $p = \lambda/(\lambda + \mu)$, so that $\lambda/\mu = p/(1 - p)$. In that parametrisation

$$\frac{1}{A_\mathrm{c}} = \frac{1-p}{1-2p}, \qquad \text{that is} \qquad A_\mathrm{c}(p) = \frac{1-2p}{1-p}, \tag{2.46}$$

the subscript marking this as the continuous-time constant, to distinguish it from the discrete-time $A(p)$ of (2.26).

Figure 2.9 plots $A_\mathrm{c}(p)$ against numerically obtained values of $A(p)$, and raises an obvious question: why does the continuous curve start at 1 and the discrete one at $\frac{1}{2}$? Merely halving A_c does not superpose them, because at the other end of the range they meet. The answer is given in section 2.6.4, once the necessary machinery is in place; it turns out that $A_\mathrm{c}(p)/2$ is an exact lower bound for $A(p)$, attained only in the limit $p \to 0$.

2.5.5 Continuous-time birth–death–catastrophe

The discrete process of section 2.5.3 has a continuous-time counterpart, and since it is the central object of several later chapters it is defined here, in the notation established above. No results about it are derived in this chapter.

Definition 2.5 (Birth–death–catastrophe process)**.** Let X_t count individuals in a compartment. Each individual independently gives birth at rate λ and dies at rate μ, and the compartment additionally undergoes a catastrophe — a rupture that removes the entire population at once —

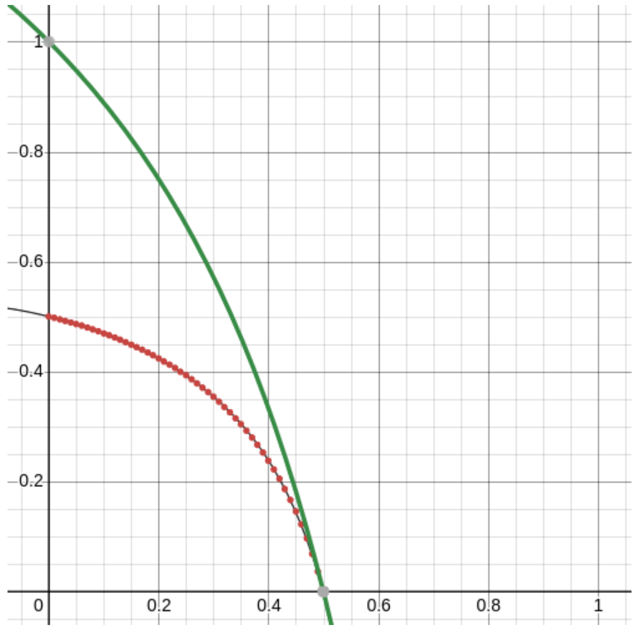


Figure 2.9: The continuous-time constant $A_{\mathrm{c}}(p) = (1-2p)/(1-p)$ (green) against high-precision numerical values of the discrete-time constant $A(p)$ (red points). Both vanish at $p = \frac{1}{2}$, but at $p = 0$ the continuous curve starts at 1 and the discrete one at $\frac{1}{2}$, and the two are not related by a constant factor in between. The black line is a formula obtained by symbolic regression (section 2.6.5); it tracks the points closely but is not exact. Section 2.6.4 accounts for the discrepancy exactly.

at a rate ρ per individual, so that the total catastrophe rate in state n is ρn. Writing † for the post-catastrophe state, the non-zero transition rates out of $n \geq 1$ are

$$q_{n,n+1} = \lambda n, \qquad q_{n,n-1} = \mu n, \qquad q_{n,\dagger} = \rho n, \tag{2.47}$$

and both $n = 0$ and † are absorbing.

Two features distinguish definition 2.5 from the linear birth–death process of section 2.2.4, and both matter later. The process now has *two* absorbing states, reached by mechanisms that are distinguishable and not interchangeable: extinction empties the compartment gradually, catastrophe empties it in one event and releases its contents. And the catastrophe rate is proportional to the population, so the compartment becomes more likely to rupture the fuller it is.

The generating-function machinery of section 2.2.3 applies unchanged. Restricting attention to realisations that have not yet ruptured and writing $G(z,t) = \mathbb{E}\Big(z^{X_t}\mathbf{1}\{X_t \neq \dagger\}\Big)$, the master equation for (2.47) yields

$$\frac{\partial G}{\partial t} = \big(\lambda z^2 - (\lambda + \mu + \rho)z + \mu\big)\frac{\partial G}{\partial z}, \qquad G(z,0) = z^N, \tag{2.48}$$

which differs from (2.13) only in the coefficient of z. The quadratic no longer factorises with a root at $z = 1$, and that single change is responsible for most of what is interesting about the process: probability is not conserved on the non-ruptured states, and $G(1,t)$ is the survival probability rather than one. Rates, means and variances, state probabilities and the associated generating-function equations are developed in the later chapters on birth–death–catastrophe processes, with the two-type extension treated in the chapter on multi-type processes and the release of a ruptured compartment's contents into a shared medium in the chapter on compartment rupture. The method used to solve (2.48) and its relatives is the subject of section 2.7.

2.6 The constant $A(p)$

Everything in section 2.4 reduced to the single constant

$$A(p) = \lim_{n\to\infty} \frac{S_n}{(2p)^n}, \qquad p \in [0, \tfrac{1}{2}), \tag{2.49}$$

whose reciprocal is the quasi-stationary mean. The results of this section are original to this thesis, and are placed in a background chapter because the quasi-stationary material above reduces to $A(p)$ and because later chapters use the resulting numerical scheme throughout.

Approximate values are easy enough to obtain: iterate $S_{n+1} = 2pS_n - pS_n^2$ until S_n is very nearly stationary, then form $S_n/(2p)^n$. Repeating over a range of p traces the curve of figure 2.9, and this works well over most of the interval. It fails as $p \to \frac{1}{2}^-$, because the number of generations needed for S_n to reach near-stationarity grows without bound.

No formula in elementary functions is known for $A(p)$, and a good deal of effort went into looking for one. This section sets out what was found: an exact infinite product (section 2.6.1); an exact series with two-sided bounds that pin $A(p)$ between the continuous-time constant of section 2.5.4 and twice it (section 2.6.2); the near-critical asymptotic $A(p) \sim 2(1-2p)$ (section 2.6.3); a resolution of the discrepancy between the discrete and continuous curves (section 2.6.4); an account of the symbolic-regression searches (section 2.6.5); and finally the dynamical argument that explains why those searches were doomed (section 2.6.6). Section 2.6.7 draws the practical conclusions.

2.6.1 An infinite-product representation

Take the nonlinear map for the survival probability,

$$S_{n+1} = 2pS_n - pS_n^2, \qquad S_0 = 1,$$

and substitute $S_n = 2w_n$. This gives

$$w_{n+1} = 2p\, w_n(1 - w_n) = r\, w_n(1 - w_n), \qquad r = 2p, \quad w_0 = \tfrac{1}{2}, \tag{2.50}$$

which is the logistic map — the same map that exhibits the period-doubling route to chaos as r increases from 0 to 4 (figure 2.10). The subcritical window is $r \in [0, 1)$, at the extreme left of that picture, where the origin is a globally attracting fixed point. That this innocuous corner of the logistic family is nevertheless enough to obstruct a closed form is the substance of section 2.6.6.

In terms of w, (2.49) reads

$$A(p) = \lim_{n\to\infty} \frac{S_n}{(2p)^n} = 2 \lim_{n\to\infty} \frac{w_n}{r^n}. \tag{2.51}$$

Any sequence can be written as a telescoping product,

$$x_n = x_0 \left(\frac{x_1}{x_0}\right)\left(\frac{x_2}{x_1}\right)\cdots\left(\frac{x_n}{x_{n-1}}\right) = x_0 \prod_{k=0}^{n-1} \frac{x_{k+1}}{x_k}, \qquad \text{so} \qquad \lim_{n\to\infty} x_n = x_0 \prod_{k=0}^{\infty} \frac{x_{k+1}}{x_k},$$

and applying this with $x_n = w_n/r^n$ the ratio simplifies at once:

$$\frac{x_{k+1}}{x_k} = \frac{w_{k+1}}{w_k} \cdot \frac{1}{r} = \frac{r w_k(1 - w_k)}{w_k} \cdot \frac{1}{r} = 1 - w_k.$$

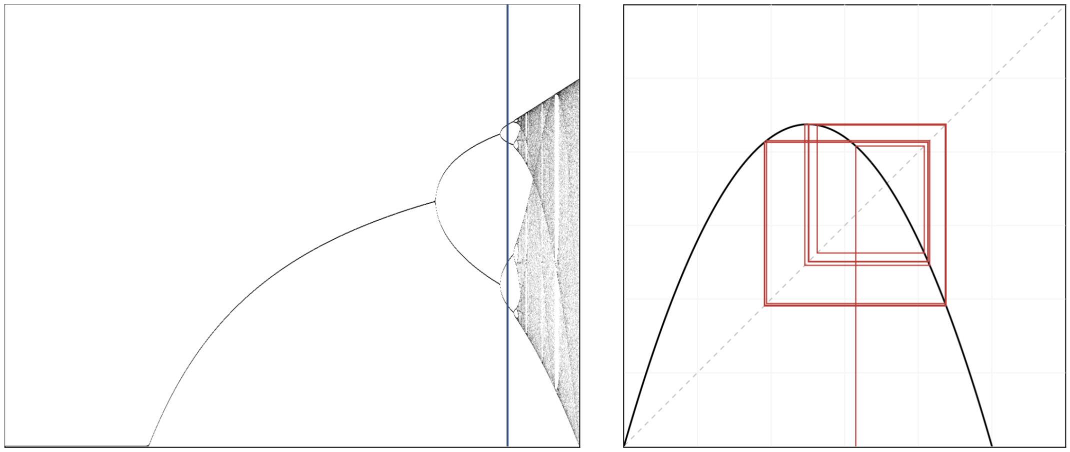


Figure 2.10: Bifurcation diagram of the logistic map $w \mapsto rw(1-w)$, showing the period doubling of stable states and the onset of chaos. The blue sample line on the left is at $r = 3.5255$; the right-hand plot shows a trajectory of the map at the same r. The regime relevant to subcritical branching is $r = 2p \in [0, 1)$, off the left-hand edge of this diagram, where the dynamics are as simple as they can be.

Hence, with $x_0 = w_0 = \frac{1}{2}$ and $A(p) = 2 \lim_n x_n$,

$$A(p) = 2 \cdot \tfrac{1}{2} \prod_{k=0}^{\infty} (1 - w_k) = \prod_{k=0}^{\infty} (1 - w_k),$$

and pulling out the factor $1 - w_0 = \frac{1}{2}$,

$$A(p) = \frac{1}{2} \prod_{k=1}^{\infty} (1 - w_k) = \frac{1}{2} \prod_{k=1}^{\infty} \left(1 - \frac{S_k}{2}\right). \tag{2.52}$$

Convergence

For $0 < w_k < 1$ the infinite product $\prod_{k \geq 1}(1 - w_k)$ converges to a strictly positive limit if and only if $\sum_{k \geq 1} w_k < \infty$. From (2.50) with $0 < r < 1$ and $0 < w_n < 1$ we have $w_{n+1} < rw_n$, so by induction $w_n \leq w_0 r^n$; since $0 < r < 1$ this bound is summable, and the product converges to a positive finite value. That also establishes what was asserted in section 2.4.1: the limit (2.49) defining $A(p)$ exists and lies strictly between 0 and $\frac{1}{2}$ for $p \in (0, \frac{1}{2})$.

Is this an advance?

Only a modest one. (2.52) is exact and its partial products decrease monotonically, so truncating gives a rigorous upper bound on $A(p)$ at every order, which the raw ratio $S_n/(2p)^n$ does not. But computing the factors still requires iterating the nonlinear map to generate the S_k, so the near-critical difficulty is untouched. What the product does supply is a route to the sharper result of the next subsection.

2.6.2 An exact series and two-sided bounds

The product converts into a telescoping sum, and the sum is much more informative. Write $r = 2p$ and note that the recursion factorises as

$$S_{n+1} = rS_n\left(1 - \frac{S_n}{2}\right).$$

Define $v_n = r^n/S_n$, so that $v_n \to 1/A(p)$ by (2.49). Then

$$v_{n+1} - v_n = \frac{r^{n+1}}{rS_n\left(1 - \frac{S_n}{2}\right)} - \frac{r^n}{S_n} = \frac{r^n}{S_n}\left[\frac{1}{1 - \frac{S_n}{2}} - 1\right] = \frac{r^n}{S_n} \cdot \frac{S_n/2}{1 - \frac{S_n}{2}} = \frac{r^n}{2 - S_n}.$$

Summing from $v_0 = 1/S_0 = 1$ gives an exact representation of the quasi-stationary mean.

Proposition 2.6 (Series representation)**.** *For $p \in [0, \frac{1}{2})$, with S_n generated by (2.22),*

$$\frac{1}{A(p)} = 1 + \sum_{n=0}^{\infty} \frac{(2p)^n}{2 - S_n}. \tag{2.53}$$

Unlike the product, this representation isolates the geometric decay in a factor that does not depend on the dynamics at all; the whole of the nonlinearity sits in the bounded denominator $2 - S_n$. That is what makes it useful, because S_n is confined to $(0, 1]$ and therefore $2 - S_n$ to $[1, 2)$. Bounding the denominator by its extremes and summing the resulting geometric series gives the following.

Proposition 2.7 (Two-sided bounds)**.** *With $\varepsilon = 1 - 2p$ as in (2.35), for $p \in (0, \frac{1}{2})$,*

$$\frac{\varepsilon}{1 + \varepsilon} \ < \ A(p) \ < \ \frac{2\varepsilon}{1 + 2\varepsilon}, \tag{2.54}$$

that is,

$$\frac{1 - 2p}{2(1 - p)} \ < \ A(p) \ < \ \frac{2(1 - 2p)}{3 - 4p}. \tag{2.55}$$

The lower bound is exactly half the continuous-time constant (2.46):

$$A(p) > \tfrac{1}{2} A_{\mathrm{c}}(p). \tag{2.56}$$

Proof. Since $0 < S_n \le 1$ for all n we have $1 \le 2 - S_n < 2$, and hence $\frac{1}{2}r^n < r^n/(2 - S_n) \le r^n$, with equality on the right only at $n = 0$, where $S_0 = 1$. Summing over $n \ge 0$ and using $\sum_n r^n = 1/(1 - r) = 1/\varepsilon$,

$$1 + \frac{1}{2\varepsilon} \ < \ \frac{1}{A(p)} \ < \ 1 + \frac{1}{\varepsilon},$$

and inverting gives (2.54). Substituting $\varepsilon = 1 - 2p$ gives (2.55), whose lower bound is $(1 - 2p)/(2(1 - p)) = \frac{1}{2}A_{\mathrm{c}}(p)$ by (2.46). □

Two features of proposition 2.7 deserve comment, because between them they account for the whole shape of the discrete curve.

At the far subcritical end, $p \to 0$, we have $\varepsilon \to 1$ and the lower bound tends to $\frac{1}{2}$. Since $A(p) \le \frac{1}{2}$ by proposition 2.8 below, the bound is asymptotically attained and $A(0^+) = \frac{1}{2}$. At the critical end, $p \to \frac{1}{2}^-$, we have $\varepsilon \to 0$, the lower bound behaves like ε and the upper like 2ε; section 2.6.3 shows that it is the upper bound that is attained there. So $A(p)$ runs from its lower bound at one end of the interval to its upper bound at the other, which is why no constant rescaling of A_{c} can reproduce it.

Proposition 2.8 (Parity bound)**.** *For the binary Galton–Watson process started from a single individual, $A(p) \le \frac{1}{2}$ for all $p \in [0, \frac{1}{2})$, with equality approached as $p \to 0$.*

Proof. Every individual leaves either 0 or 2 offspring, so Z_n is even for every $n \ge 1$. Conditional on survival, therefore, $Z_n \ge 2$, and hence $\mathbb{E}(Z_n \mid Z_n > 0) \ge 2$ for all $n \ge 1$. Letting $n \to \infty$ and using (2.28) gives $1/A(p) \ge 2$. As $p \to 0$ the conditional population is 2 with probability tending to one, so the bound is attained in the limit. □

Table 2.1: The constant $A(p)$ computed from the product (2.52) in 60-digit arithmetic, against the bounds of proposition 2.7 and the near-critical asymptotic $2(1-2p)$ of section 2.6.3. The bounds are loose in the middle of the range and tighten at both ends: at $p = 0.05$ the lower bound is already within 3% of A, whereas at $p = 0.4999$ it is the upper bound that is within 0.1%. The last two columns give the quasi-stationary mean and the number of factors of (2.52) needed for the product to reach full precision — the cost that section 2.6.7 has to manage.

p	$\frac{1}{2}A_{\mathrm{c}}(p)$	$A(p)$	$\dfrac{2(1-2p)}{3-4p}$	$2(1-2p)$	$1/A(p)$	iterations
0.05	0.473684	0.486180	0.642857	1.800000	2.0568	45
0.1	0.444444	0.469382	0.615385	1.600000	2.1305	64
0.2	0.375000	0.423895	0.545455	1.200000	2.3591	113
0.3	0.285714	0.353967	0.444444	0.800000	2.8251	201
0.4	0.166667	0.237647	0.285714	0.400000	4.2079	458
0.45	0.090909	0.145069	0.166667	0.200000	6.8933	966
0.48	0.038462	0.068011	0.074074	0.080000	14.7036	2 473
0.49	0.019608	0.036395	0.038462	0.040000	27.4764	4 965
0.499	0.001996	0.003944	0.003984	0.004000	253.519	48 992
0.4999	0.000200	0.000399	0.000400	0.000400	2504.65	478 905

2.6.3 Behaviour as $p \to \frac{1}{2}^{-}$

The series (2.53) also delivers the near-critical asymptotic, and does so more transparently than the matching argument one is first tempted to try.

Fix ε small. The summand $r^n/(2-S_n)$ involves two decays on very different scales. The geometric factor $r^n = (1-\varepsilon)^n$ decays on the scale $n \sim 1/\varepsilon$, which is large. The survival probability S_n, by contrast, is close to its critical behaviour $S_n \sim 2/n$ over that whole range, and so has already fallen to near zero after $O(1)$ generations. For the overwhelming majority of the terms that carry weight, therefore, $S_n \approx 0$ and the summand is approximately $r^n/2$. Summing,

$$\sum_{n=0}^{\infty} \frac{r^n}{2 - S_n} \;\approx\; \frac{1}{2}\sum_{n=0}^{\infty} r^n = \frac{1}{2\varepsilon},$$

so $1/A(p) \sim 1/(2\varepsilon)$ and

$$\boxed{A(p) \sim 2\varepsilon = 2(1-2p) \qquad \text{as } p \to \tfrac{1}{2}^{-}.} \tag{2.57}$$

Equivalently $1/A(p) \sim 1/(2(1-2p))$: the quasi-stationary mean diverges like the reciprocal of the distance from criticality. In particular $A(p)$ meets the axis at $p = \frac{1}{2}$ with gradient -4, a fact used repeatedly below as a test of candidate formulae.

The same result can be reached through the continuum equation of section 2.4.4, and it is worth seeing why that route is the worse one. The exact difference is

$$S_{n+1} - S_n = (2p-1)S_n - pS_n^2 = -\varepsilon S_n - \tfrac{1}{2}S_n^2 + \tfrac{\varepsilon}{2}S_n^2,$$

and dropping the $O(\varepsilon S_n^2)$ term gives $\mathrm{d}S/\mathrm{d}n = -\varepsilon S - \frac{1}{2}S^2$. Setting $u = 1/S$ linearises this to $u' = \varepsilon u + \frac{1}{2}$, with solution

$$u(n) = C\mathrm{e}^{\varepsilon n} - \frac{1}{2\varepsilon}, \qquad S(n) = \frac{1}{C\mathrm{e}^{\varepsilon n} - \frac{1}{2\varepsilon}}.$$

Matching $S_n \sim A\mathrm{e}^{-\varepsilon n}$ for small ε identifies $A = 1/C$, and matching to the critical decay $S_n \sim 2/n$ in the appropriate double limit fixes $C \sim 1/(2\varepsilon)$, recovering (2.57). Both steps are heuristic, and the second is delicate; the series argument is preferable because it needs no second matching at all.

Remark 2.9 (Rate of approach). Numerically, $A(p)/2\varepsilon$ approaches 1 rather slowly. Evaluating (2.52) in high precision gives $A/2\varepsilon = 0.90987$, 0.98612, 0.99814, 0.99977 and 0.999972 at $\varepsilon = 2\times10^{-2}$ down to 2×10^{-6}. Fitting the deficit gives $1 - A/2\varepsilon \approx \varepsilon\big(\ln(1/\varepsilon) + c\big)$ with $c \approx 0.78$, so

$$A(p) = 2\varepsilon\Big[1 + O\big(\varepsilon\ln(1/\varepsilon)\big)\Big].$$

The logarithmic correction is why (2.57) is a good practical surrogate only very close to criticality, and it is what fixes the crossover point chosen in section 2.6.7.

2.6.4 Why the discrete and continuous constants differ

The question raised by figure 2.9 can now be settled. The two constants are

$$A_{\mathrm{c}}(p) = \frac{1-2p}{1-p} \quad \text{(continuous time)}, \qquad A(p) = \lim_{n\to\infty}\frac{S_n}{(2p)^n} \quad \text{(discrete time)},$$

and the observation was that the first starts at 1 where the second starts at $\frac{1}{2}$, yet halving the first does not superpose the curves.

Proposition 2.7 explains this completely. What holds is not $A = \frac{1}{2}A_{\mathrm{c}}$ but the strict inequality $A > \frac{1}{2}A_{\mathrm{c}}$, and the inequality is attained only in the limit $p \to 0$. Combining (2.46) and (2.57), the ratio runs monotonically across the interval:

$$\frac{A(p)}{A_{\mathrm{c}}(p)} \longrightarrow \tfrac{1}{2} \quad (p\to 0), \qquad \frac{A(p)}{A_{\mathrm{c}}(p)} \longrightarrow 1 \quad (p \to \tfrac{1}{2}^-), \tag{2.58}$$

taking the values 0.503, 0.528, 0.619, 0.798, 0.928 and 0.988 at $p = 0.01$, 0.1, 0.3, 0.45, 0.49, 0.499. No constant factor will do, because the ratio is not constant.

The two endpoints have clean interpretations, and they are different in kind.

At $p \to 0$ the factor of two is a counting effect. In the binary discrete model an individual leaves either nothing or two offspring, so a surviving population is always even (proposition 2.8); deep in the subcritical regime, conditioning on survival means conditioning on a population of exactly two, and the quasi-stationary mean is 2. In the continuous-time model a survivor is a single individual that has not yet died, so the quasi-stationary mean is 1. The reciprocals are $\frac{1}{2}$ and 1, and there is the discrepancy.

At $p \to \frac{1}{2}^-$ the factor disappears, because both constants are governed by the same critical scaling. Expanding (2.46) near $p = \frac{1}{2}$ gives $A_{\mathrm{c}}(p) = (1-2p)/(1-p) \to 2(1-2p)$, which is precisely

(2.57). Near criticality the discreteness of the generational clock stops mattering; what survives is the universal 2ε behaviour, and the two models agree to leading order.

The discrete curve is therefore squeezed between $\frac{1}{2}A_\mathrm{c}$ and its own critical asymptote, hugging the former at one end and the latter at the other. That is the whole content of figure 2.9.

2.6.5 Searching for a closed form

Before the connection of section 2.6.6 was appreciated, several methods were tried in the hope of an elementary formula for $A(p)$. The negative result is worth recording, and so is the manner of the failure.

The first method was the continuum route of section 2.4.4: treat the map as a differential equation in the large-generation limit and solve it. This gives the shape of the decay but leaves the constant of integration undetermined, which is to say it leaves $A(p)$ undetermined.

The second was symbolic regression by genetic algorithm. The idea is simple: generate fifty candidate functions at random, measure how closely each reproduces the target curve, retain the best five, and build a fresh population of fifty by varying those; then measure again, and repeat the cycle of variation and selection many times over [12, 13]. The first implementation searched a space of polynomials, on the hunch that the answer might take that form and because such functions are simple to encode: a genome is a coefficient list $[a_0, \ldots, a_k]$ for $a_0 + a_1 p + \cdots + a_k p^k$ up to some maximum order, drawn uniformly at random. It became apparent quickly that many substantially different formulae produce curves indistinguishable from the target by eye. Supposing that a true formula might have rational coefficients, a second version searched $\sum_i (a_i/b_i)p^i$ with $[a_i], [b_i] \in \mathbb{Z}^{k+1}$, and a third extended the space to ratios of two such polynomials. The best that emerged merely fitted the high-precision numerical data well:

$$\hat{A}_1(p) = \frac{\frac{1}{2}(1-2p)(1+2p)}{1 + \frac{1}{2}p - \frac{5}{2}p^2 - \frac{6}{5}p^3 + \frac{6}{5}p^4}. \tag{2.59}$$

This passes through $(0, \frac{1}{2})$ and $(\frac{1}{2}, 0)$ and looks convincing, but it fails the diagnostic of (2.57): differentiating at $p = \frac{1}{2}$ gives a gradient of $-2/0.55 = -3.636$ rather than the required -4.

A later attempt used a symbolic tree search over progressively nested elementary functions of all kinds. The conclusion was the same. Many functions of widely varying form resemble the target closely, and no satisfactory simple formula emerges even when candidates with fewer components are explicitly favoured. The expressions returned were typically deep nestings of trigonometric and iterated exponential terms carrying a dozen or more fitted decimal constants, and they were not interpretable in any useful sense; a representative and comparatively concise one is

$$\hat{A}_3(p) = \frac{-0.616647881739505\,\mathrm{e}^{-0.552605335919693\,\mathrm{e}^{p}}}{\sin(\cos p) - |p|} + 0.921964323679771. \tag{2.60}$$

The recurring defect is the same in every case. The candidates either fail to pass through $(\frac{1}{2}, 0)$ or fail to arrive there with gradient -4, and near criticality this matters enormously: as $A \to 0$, an error in A of vanishing absolute size produces an $O(1)$ — eventually unbounded — error in the quasi-stationary mean $1/A$. A formula can look perfect on a plot of A and be useless on a plot of $1/A$.

A pragmatic patch is to splice the asymptotic onto a fitted curve,

$$\hat{A}_4(p) = \begin{cases} -0.04011483983619545\,\mathrm{e}^{\mathrm{e}^{2p}} + 0.6079994336528085, & p < 0.499962, \\ 2 - 4p, & p \geq 0.499962, \end{cases} \tag{2.61}$$

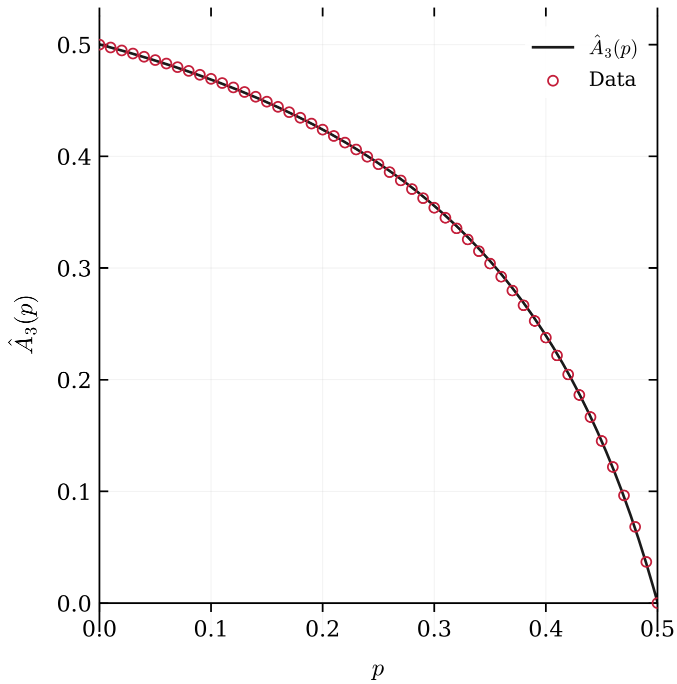


Figure 2.11: The symbolic-regression candidate $\hat{A}_3(p)$ of (2.60) (solid) against numerically computed values of $A(p)$ (red circles). The agreement is excellent to the eye across the whole interval, which is exactly the problem: the fit is not evidence of correctness. $\hat{A}_3$ misses the critical point, returning $\hat{A}_3(\frac{1}{2}) = 9.1 \times 10^{-4}$ rather than 0, and arrives there with gradient -3.63 instead of -4. Errors of this size are invisible here but are fatal for $1/A$ near criticality.

and this does work. But if one is going to splice, it is better to splice onto something exact, and section 2.6.7 does so.

The deeper lesson points directly at the next subsection. That an enormous number of very different formulae can behave almost identically over a bounded interval is not merely an inconvenience of the method; it is a symptom. Curve fitting cannot distinguish a function that has a closed form from one that does not.

2.6.6 The Koenigs connection

Some time after the search had been abandoned, a dynamical argument emerged that explains why no elementary closed form should be expected. In short: $A(p)$ is a value of the Koenigs linearising function of the logistic map, and that function is known to be non-elementary throughout the parameter range required.

Definition 2.10 (Koenigs function)**.** Let $f(z) = rz(1-z)$ have a fixed point at $z = 0$ with multiplier $r = f'(0)$, and suppose $|r| \notin \{0, 1\}$. Koenigs [14] proved that there is a unique function ψ_r, analytic near the origin, satisfying the Schröder equation

$$\psi_r(f(z)) = r\,\psi_r(z), \qquad \psi_r(0) = 0, \qquad \psi_r'(0) = 1. \tag{2.62}$$

The Koenigs function is a change of coordinate that straightens the dynamics: in the ψ coordinate the nonlinear map f becomes multiplication by r (figure 2.12). Here $f(z) = rz(1-z)$ is exactly

(2.50), with $z = w_n$ and $f(z) = w_{n+1}$, and the multiplier is $r = 2p \in (0, 1)$ — an attracting fixed point, which is the classical Koenigs setting. Standard accounts are Milnor [15] and Carleson and Gamelin [16].

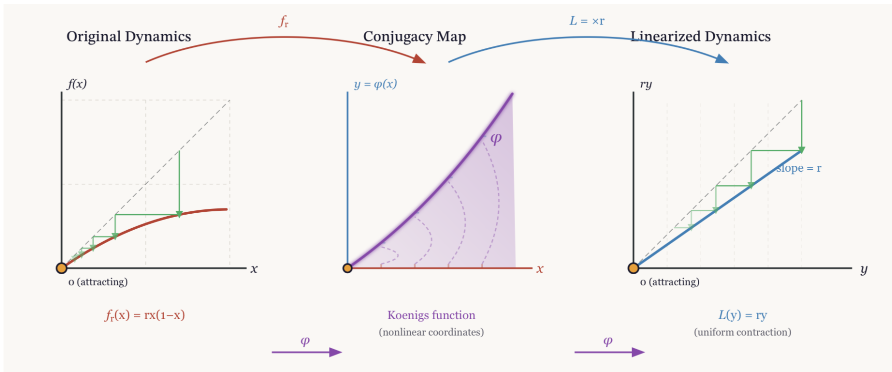


Figure 2.12: Koenigs linearisation of the logistic map near an attracting fixed point. The nonlinear dynamics $x \mapsto f_r(x) = rx(1-x)$ are conjugated by ψ_r to the linear map $y \mapsto ry$; the diagram displays the functional relation $\psi_r(f_r(x)) = r\psi_r(x)$ and the role of ψ_r as the coordinate change that straightens the dynamics.

Proposition 2.11. *For $p \in (0, \frac{1}{2})$ and $r = 2p$,*

$$A(p) = 2\,\psi_r\left(\tfrac{1}{2}\right). \tag{2.63}$$

Proof. Iterating (2.62) along the orbit $w_0, w_1, w_2, \ldots$ of (2.50) gives $\psi(w_1) = r\psi(w_0)$, hence $\psi(w_0) = \psi(w_1)/r$; similarly $\psi(w_1) = \psi(w_2)/r$, so $\psi(w_0) = \psi(w_2)/r^2$; and by induction

$$\psi(w_0) = \frac{\psi(w_n)}{r^n} \qquad \text{for every } n. \tag{2.64}$$

Since (2.64) holds for all n we may pass to the limit, $\psi(w_0) = \lim_n \psi(w_n)/r^n$. The normalisation $\psi(0) = 0$, $\psi'(0) = 1$ means $\psi(z)/z \to 1$ as $z \to 0$, and $w_n = S_n/2 \to 0$ because $r < 1$; replacing $\psi(w_n)$ by w_n in the limit is therefore legitimate, and $\psi(w_0) = \lim_n w_n/r^n$. Comparing with (2.51) and using $w_0 = S_0/2 = \frac{1}{2}$ gives $A(p) = 2\psi_r(\frac{1}{2})$. ☐

When is ψ_r elementary?

Essentially never. The logistic family is affinely conjugate to the canonical quadratic family $P_c(z) = z^2 + c$; substituting $x = -z/r + \frac{1}{2}$ in f_r gives P_c with

$$c = \frac{2r - r^2}{4}, \tag{2.65}$$

under which the interval $r \in (0, 1)$ maps to $c \in (0, \frac{1}{4})$ (figure 2.13). The relevance of this is a rigidity theorem of Becker and Bergweiler [17]: for a rational map R of degree at least two with a fixed point of multiplier λ, $0 < |\lambda| < 1$, the associated Koenigs linearisation satisfies an algebraic differential equation only when R is analytically conjugate to a monomial map $z \mapsto z^{\pm d}$ or to a Chebyshev polynomial $\pm T_d$. For the quadratic family those exceptional cases are exactly $c = 0$ and $c = -2$, corresponding under (2.65) to $r = 2$ and $r = 4$.

Every elementary function in the sense of Liouville — algebraic functions, exponentials, logarithms and their finite combinations — satisfies an algebraic differential equation, so a function satisfying none, a *hypertranscendental* function, cannot be elementary. And

$$\left(0, \tfrac{1}{4}\right) \cap \{0, -2\} = \emptyset,$$

so no r in the branching window is exceptional: for each fixed $r \in (0, 1)$ the Koenigs function ψ_r is hypertranscendental, and in particular not elementary.

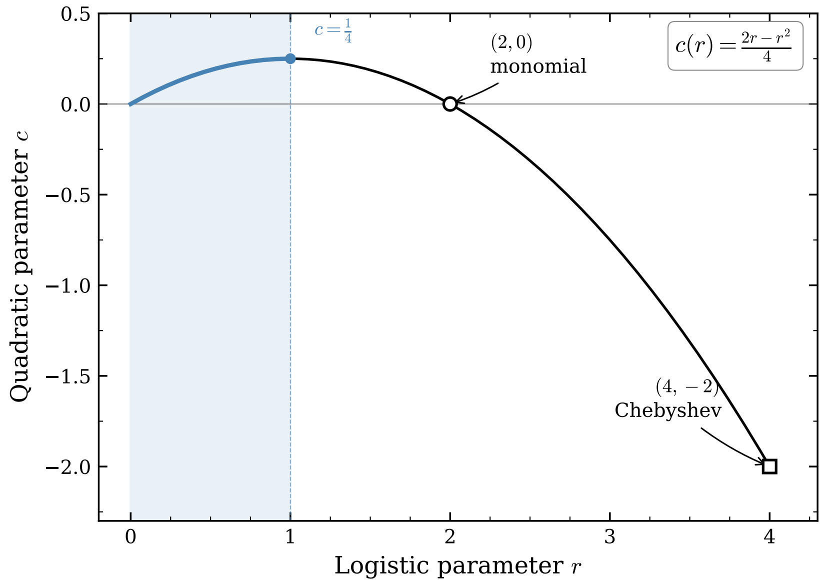


Figure 2.13: The affine parameter correspondence $c(r) = (2r - r^2)/4$ between the logistic family $f_r(x) = rx(1 - x)$ and the quadratic family $P_c(z) = z^2 + c$. The shaded segment is $r \in (0, 1)$, the subcritical branching window, which maps to $c \in (0, \frac{1}{4})$. The two integrable parameters, $r = 2$ ($c = 0$, monomial) and $r = 4$ ($c = -2$, Chebyshev), lie outside it.

The two exceptional parameters are worth writing out, both as a check on the machinery and because they show how special the solvable cases are.

The monomial case, $r = 2$**.** Here $f_2(x) = 2x(1 - x)$, and the substitution $u = 1 - 2x$ gives

$$u_{n+1} = 1 - 2\big(2x_n(1 - x_n)\big) = 1 - 4x_n + 4x_n^2 = (1 - 2x_n)^2 = u_n^2,$$

a map linearised by the logarithm, since $\ln(u^2) = 2 \ln u$. Returning to x and imposing the normalisation of definition 2.10,

$$\psi_2(x) = -\tfrac{1}{2} \ln(1 - 2x), \tag{2.66}$$

which is readily verified: $\psi_2(f_2(x)) = -\frac{1}{2} \ln\big((1 - 2x)^2\big) = 2\psi_2(x)$.

The Chebyshev case, $r = 4$**.** Here $f_4(x) = 4x(1 - x)$ is conjugate to $T_2(z) = 2z^2 - 1$ by a trigonometric substitution. Putting $x = \sin^2 \theta$,

$$f_4(x) = 4 \sin^2 \theta \cos^2 \theta = (2 \sin \theta \cos \theta)^2 = \sin^2(2\theta),$$

so the map is a doubling $\theta \mapsto 2\theta$ in the θ coordinate. The multiplier at the origin is 4 rather than 2, which suggests that the linearisation involves θ^2; and indeed

$$\psi_4(x) = \left(\arcsin\sqrt{x}\right)^2. \tag{2.67}$$

Using $\arcsin(2u\sqrt{1-u^2}) = 2\arcsin u$ with $u = \sqrt{x}$ gives $\psi_4(f_4(x)) = (2\arcsin\sqrt{x})^2 = 4\psi_4(x)$; and as $x \to 0$, $\arcsin\sqrt{x} \approx \sqrt{x}$, so $\psi_4(x) \approx x$ and $\psi_4'(0) = 1$.

Remark 2.12 (Attracting versus repelling). At $r = 2$ and $r = 4$ the multiplier at the origin has $|r| > 1$, so the origin is *repelling*. The functions (2.66) and (2.67) are therefore Poincaré-type linearisations rather than attracting-Koenigs maps, and neither parameter lies in the branching window $r = 2p \in [0, 1)$ in any case. They are recorded to show what a solvable case looks like, not because they bear on $A(p)$.

What this does and does not establish

It is worth being precise about the status of the conclusion, because it is easy to overstate.

What is established is that for each *fixed* $r \in (0, 1)$ the function $z \mapsto \psi_r(z)$ is hypertranscendental. Combined with proposition 2.11, this says that $A(p)$ is a value of a non-elementary function, evaluated at the fixed argument $z = \frac{1}{2}$, for every p in the subcritical range. It is a strong indication that the search of section 2.6.5 was doomed, and it explains the phenomenology of that search: the object being fitted is defined only by the dynamics that generate it.

What is *not* established is that the map $p \mapsto A(p)$ admits no elementary description. Hypertranscendence of ψ_r in its argument z is a statement about one variable, whereas $A(p) = 2\psi_{2p}(\frac{1}{2})$ varies the parameter and freezes the argument; the two questions are logically distinct. A rigorous negative result for $p \mapsto A(p)$ would require a transcendence statement about the values rather than the functions, and none is claimed here. The honest summary is that no closed form is known, that the structural reason for expecting none is the one just given, and that the numerical evidence of section 2.6.5 is entirely consistent with there being none.

It is worth seeing where in the quadratic family these parameters sit, because the picture makes the difficulty look structural rather than accidental. The main cardioid of the Mandelbrot set — the set of c for which $z^2 + c$ has an attracting fixed point — is parametrised by that point's multiplier λ through $c = \lambda/2 - \lambda^2/4$, which is exactly (2.65) with $\lambda = r$. The branching window $r \in (0, 1)$ is therefore the real diameter of the main cardioid, traversed from the centre $c = 0$ out to the parabolic cusp $c = \frac{1}{4}$, where $\lambda = 1$ and the process turns critical. Figure 2.14 places the segment in context.

Throughout the window the Julia set is a quasicircle bounding the basin of the attracting fixed point, and it is a genuine fractal for every c in the interval. The Koenigs function is the conformal coordinate that linearises the dynamics inside that basin, so its natural domain is bounded by exactly this curve. A function whose domain of definition has no smooth description is not a promising candidate for a closed form, and $A(p)$, being one of its values, inherits the difficulty. The contrast with the integrable cases is sharp: at $c = 0$ the Julia set degenerates to the unit circle, and it is precisely there that the linearisation becomes elementary, (2.66).

2.6.7 Computing $A(p)$ in practice

In the absence of an exact formula we are left with approximation, and the two regimes of section 2.6.2 suggest how to split the work.

Over most of the interval, direct iteration is entirely satisfactory: the infinite product (2.52) converges quickly, its partial products bound $A(p)$ from above at every order, and modest numbers

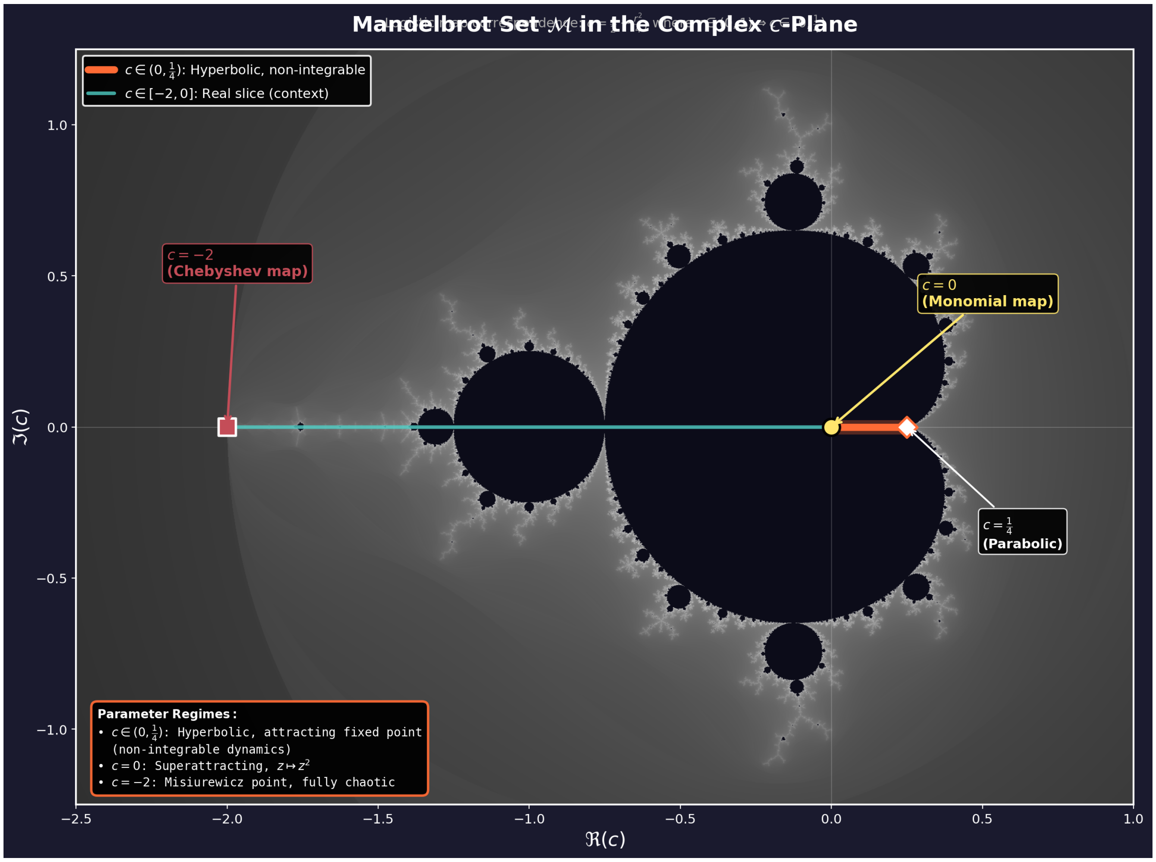


Figure 2.14: The Mandelbrot set in the complex c-plane, with the interval $c \in (0, \frac{1}{4})$ highlighted. Under the correspondence $c = (2r - r^2)/4$ of (2.65) this is the image of the subcritical branching window $r = 2p \in (0, 1)$, and it forms the real diameter of the main cardioid, running from the centre $c = 0$ to the parabolic cusp $c = \frac{1}{4}$, which is criticality. The two integrable parameters are marked: $c = 0$ (monomial, reached in logistic coordinates at $r = 2$) and $c = -2$ (Chebyshev, $r = 4$), the latter at the far tip of the real slice. Note that $c(r) = c(2 - r)$, so each $c \in (0, \frac{1}{4})$ is the image of two logistic parameters, r and $2 - r$; these correspond to the two fixed points of $z^2 + c$, with multipliers r and $2 - r$, and it is the attracting one, $r < 1$, that governs $A(p)$.

of iterations give many digits. The difficulty is confined to a neighbourhood of $p = \frac{1}{2}$, and it has two parts. First, as $A(p) \to 0$ the quantity of interest $1/A$ diverges, so a fixed absolute error in A becomes an unbounded error in the quasi-stationary mean. Second, and this is the binding constraint, as the process is defined ever closer to criticality its lifetime, though still finite, grows enormously, and S_n recedes to zero ever more slowly. The iteration must be carried to a generation at which S_n is essentially unchanging, and table 2.1 shows what that costs: about 10^2 iterations at $p = 0.2$, 10^3 at $p = 0.45$, but nearly 5×10^5 at $p = 0.4999$.

The remedy is to hand the near-critical region to the asymptotic (2.57), which is exact in the limit and cheap everywhere:

$$A(p) \approx \begin{cases} \dfrac{1}{2} \displaystyle\prod_{k=1}^{N} \left(1 - \dfrac{S_k}{2}\right), & p < p_*, \quad N \gg 1, \\ 2 - 4p, & p \geq p_*, \end{cases} \tag{2.68}$$

with the crossover p_* chosen just below $\frac{1}{2}$. Remark 2.9 indicates where to put it. The relative error of the asymptotic is about $\varepsilon \ln(1/\varepsilon)$, so at $p_* = 0.4999$ ($\varepsilon = 2 \times 10^{-4}$) the asymptotic is already accurate to about two parts in 10^3, while the product still converges in a few hundred thousand iterations. Any p_* in the range 0.499 to 0.4999 gives a scheme accurate to better than one per cent throughout, and the two branches can be blended over a narrow window if continuity of the derivative matters.

This is the recipe used for the numerical values quoted throughout this chapter, including those in figure 2.5 and table 2.1.

2.7 Absorption models and the method of characteristics

The models considered so far describe the population inside a single compartment. Once several compartments share a medium, a further question arises: what becomes of the contents of the first compartment to rupture? The shared medium acquires, at a stroke, a potentially large cohort of particles available for subsequent absorption by the compartments that remain.

The analysis to this point has assumed *immediate transfer*, in which the entire contents of a ruptured compartment relocate at once into the interior of another. That is a strong assumption. The alternative is an *absorption period*: following a rupture, absorption events redistribute the released cohort among the remaining compartments, and only when every particle has been taken up do birth, death and rupture resume. Working towards models of that kind, this section treats the case in which a cohort of particles enters a single compartment.

Three models are treated in increasing order of difficulty. In the first two the particles are conserved or only lost, the state space is finite, and an independence argument delivers the generating function with no need to solve a partial differential equation at all. In the third, replication inside the compartment opens an infinite state space and the independence shortcut fails, so the equation must be confronted directly. The method of characteristics is therefore developed on the second model, where the answer is already known, and then applied to the third. The closed form obtained for the third model, proposition 2.13, is a result of this thesis.

2.7.1 Absorption only

The simplest absorption process has a single compartment that takes in particles, with no subsequent dynamics: no birth and no death within the compartment.

Let X_t and Y_t count the particles inside and outside the compartment respectively; the same labelling is used for every version of the single-compartment model below. Events are stochastic with exponentially distributed inter-event times, which by section 2.2.1 follows from the assumption that event rates do not depend on time, so that $\{(X_t, Y_t) : t \geq 0\}$ is a continuous-time Markov chain in the sense of section 2.2.2.

With only absorption to consider, the model is specified by the transition probabilities over an infinitesimal interval Δt,

$$\mathbb{P}(\Delta X = 1,\ \Delta Y = -1 \mid Y_t = j) = j\alpha\,\Delta t, \qquad \text{(absorption)}$$
$$\mathbb{P}(\Delta X = 0,\ \Delta Y = 0 \mid Y_t = j) = 1 - j\alpha\,\Delta t, \qquad \text{(no event)}$$

where $\Delta X = X_{t+\Delta t} - X_t$ and similarly for Y, and terms of $o(\Delta t)$ are suppressed throughout. The factor j records that each exterior particle is independently subject to the same probability $\alpha\Delta t$

of being taken up. The initial condition is $X_0 = 0$, $Y_0 = N$: a starter population of N particles occupying the exterior medium.

Writing $p_{i,j}(t) = \mathbb{P}(X_t = i,\ Y_t = j)$, the forward Kolmogorov equations are

$$\frac{\mathrm{d}p_{i,j}}{\mathrm{d}t} = \alpha(j+1)p_{i-1,j+1} - \alpha j\, p_{i,j}. \tag{2.69}$$

With the bivariate generating function

$$G(x,y,t) = \sum_i \sum_j p_{i,j}(t)\, x^i y^j = \mathbb{E}\Big(x^{X_t} y^{Y_t}\Big)\,, \tag{2.70}$$

(2.69) converts into a partial differential equation by the procedure of section 2.2.3. Multiply each equation by $x^i y^j$ and sum over i and j; the sums are finite, because $p_{i,j}(t) = 0$ whenever $i + j \neq N$, particles in this model being neither created nor destroyed. Shifting indices in the first term sends $\alpha(j+1)p_{i-1,j+1}x^i y^j$ to $\alpha x\, \partial_y G$, and the second term is $-\alpha y\, \partial_y G$, so that

$$\frac{\partial G}{\partial t} = \alpha(x-y)\frac{\partial G}{\partial y}, \qquad G(x,y,0) = y^N. \tag{2.71}$$

This first-order linear initial value problem is more awkward than it looks. The method of characteristics applies in principle to equations in three variables, but applying it here is not straightforward, and it is postponed to section 2.7.3. The solution can be obtained by other means, and it is worth doing so both because the answer is needed and because it provides a check on the machinery developed there.

Exploiting independence

The function G above was specific to $X_0 = 0$, $Y_0 = N$. Define more generally

$$G_n(x,y,t) = \sum_i \sum_j \mathbb{P}(X_t = i,\ Y_t = j \mid X_0 = 0,\ Y_0 = n)\ x^i y^j.$$

In a process of pure absorption the n particles are taken up one by one, and the behaviour of any one of them does not depend on the movements of its neighbours. That independence lets X_t and Y_t decompose into sums of n single-particle variables,

$$X_t = \sum_{k=1}^{n} X_t^{(k)}, \qquad Y_t = \sum_{k=1}^{n} Y_t^{(k)},$$

from which

$$G_n(x,y,t) = \big(G_1(x,y,t)\big)^n, \tag{2.72}$$

since

$$G_n(x,y,t) = \mathbb{E}\left(\prod_{k=1}^{n} x^{X_t^{(k)}} \prod_{k=1}^{n} y^{Y_t^{(k)}}\right) = \prod_{k=1}^{n} \mathbb{E}\Big(x^{X_t^{(k)}} y^{Y_t^{(k)}}\Big) = \big(G_1(x,y,t)\big)^n.$$

Now consider a single initial particle, $X_0 = 0$, $Y_0 = 1$. It is eventually absorbed, and that is the final state of the system: no other states are reachable, so $p_{i,j}(t) = 0$ whenever $i + j \neq 1$. The forward equations reduce to

$$\frac{\mathrm{d}p_{0,1}}{\mathrm{d}t} = -\alpha p_{0,1}, \qquad \frac{\mathrm{d}p_{1,0}}{\mathrm{d}t} = \alpha p_{0,1},$$

giving $p_{0,1}(t) = \mathrm{e}^{-\alpha t}$ and $p_{1,0}(t) = 1 - \mathrm{e}^{-\alpha t}$ (figure 2.15). These being the only two non-zero state probabilities of the single-particle system,

$$G_1(x, y, t) = p_{1,0}(t)\, x + p_{0,1}(t)\, y = \left(1 - \mathrm{e}^{-\alpha t}\right) x + \mathrm{e}^{-\alpha t} y,$$

and (2.72) gives the generating function for any initial census,

$$G_n(x, y, t) = \left(\left(1 - \mathrm{e}^{-\alpha t}\right) x + \mathrm{e}^{-\alpha t} y\right)^n. \tag{2.73}$$

This satisfies (2.71) together with the initial condition $G_n(x, y, 0) = y^n$. The form is recognisable: $X_t \sim \text{Binomial}(n, 1 - \mathrm{e}^{-\alpha t})$, as it must be, since the n particles are absorbed independently and each has been taken up by time t with probability $1 - \mathrm{e}^{-\alpha t}$.

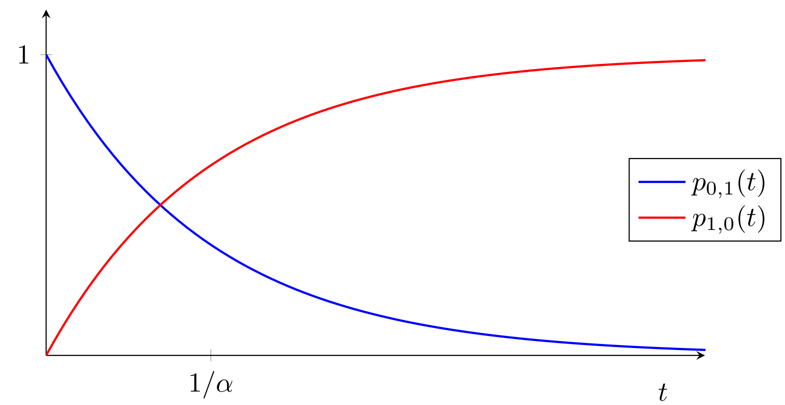


Figure 2.15: State probabilities for the single-particle absorption-only model. The particle begins outside the compartment ($p_{0,1}$, decaying exponentially at rate α) and ends inside it ($p_{1,0}$). There is nowhere else for it to go, so the two curves sum to one at every time.

2.7.2 Absorption–death

A particle resident in a compartment may die. Keeping X_t and Y_t for the interior and exterior counts, death enters as follows:

$$\begin{aligned}
\mathbb{P}(\Delta X = 1,\ \Delta Y = -1 \mid Y_t = j) &= j\alpha\, \Delta t, && \text{(absorption)}\\
\mathbb{P}(\Delta X = -1,\ \Delta Y = 0 \mid X_t = i) &= i\mu\, \Delta t, && \text{(death)}\\
\mathbb{P}(\Delta X = 0,\ \Delta Y = 0 \mid X_t = i,\ Y_t = j) &= 1 - (j\alpha + i\mu)\Delta t, && \text{(no event)}
\end{aligned}$$

with forward Kolmogorov equations

$$\frac{\mathrm{d}p_{i,j}}{\mathrm{d}t} = \alpha(j+1)p_{i-1,j+1} + \mu(i+1)p_{i+1,j} - (\alpha j + \mu i)p_{i,j}. \tag{2.74}$$

The derivation of section 2.7.1 carries over unchanged and gives the first-order linear equation

$$\frac{\partial G}{\partial t} = \mu(1-x)\frac{\partial G}{\partial x} + \alpha(x-y)\frac{\partial G}{\partial y}, \qquad G(x, y, 0) = y^N, \tag{2.75}$$

and the same difficulty. Once again the solution can be constructed via (2.72), since individual particles remain independent.

Without birth events the number of particles cannot exceed its initial value, so for an isolated particle the generating function has just three terms,

$$G_1(x, y, t) = p_{0,0}(t) + p_{1,0}(t)\, x + p_{0,1}(t)\, y.$$

The state probabilities follow from

$$\frac{\mathrm{d}p_{0,1}}{\mathrm{d}t} = -\alpha p_{0,1}, \qquad \frac{\mathrm{d}p_{1,0}}{\mathrm{d}t} = \alpha p_{0,1} - \mu p_{1,0},$$

whence

$$p_{0,1}(t) = \mathrm{e}^{-\alpha t}, \qquad p_{1,0}(t) = \frac{\alpha}{\mu - \alpha}\left(\mathrm{e}^{-\alpha t} - \mathrm{e}^{-\mu t}\right), \qquad p_{0,0}(t) = 1 - p_{0,1}(t) - p_{1,0}(t), \tag{2.76}$$

plotted in figure 2.16. Applying (2.72) gives the general generating function

$$G_n(x, y, t) = \left(1 - \tfrac{\mu}{\mu-\alpha}\mathrm{e}^{-\alpha t} + \tfrac{\alpha}{\mu-\alpha}\mathrm{e}^{-\mu t} + \tfrac{\alpha}{\mu-\alpha}(\mathrm{e}^{-\alpha t} - \mathrm{e}^{-\mu t})x + \mathrm{e}^{-\alpha t} y\right)^n. \tag{2.77}$$

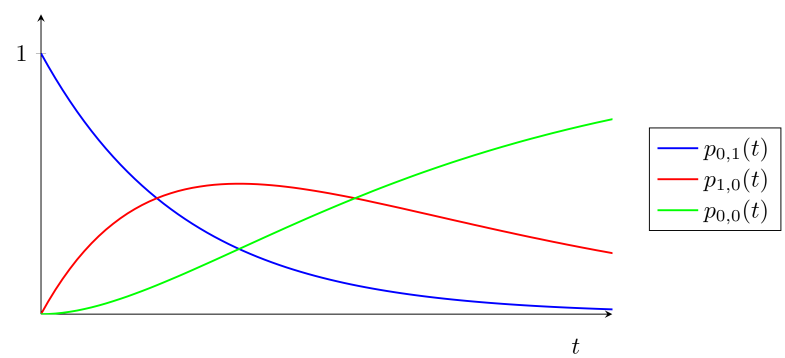


Figure 2.16: State probabilities for the single-particle absorption–death model, (2.76). The particle starts outside ($p_{0,1}$, blue), is absorbed into the compartment ($p_{1,0}$, red, which rises and then decays as death takes over), and finally dies ($p_{0,0}$, green, absorbing).

2.7.3 The method of characteristics, tested on absorption–death

Before continuing to absorption–birth–death, where replication may occur inside the compartment and the independence shortcut is unavailable, it is worth revisiting absorption–death from the point of view of its partial differential equation. Having (2.77) already in hand makes it an ideal test case.[1]

Introduce a change of variables $(x, y, t) \to (\sigma, \eta, \tau)$, writing

$$G(x, y, t) = G\big(x(\sigma, \eta, \tau),\, y(\sigma, \eta, \tau),\, t(\sigma, \eta, \tau)\big) \equiv G(\sigma, \eta, \tau).$$

The method seeks solutions along the characteristic curves of constant σ and η, on which G is a function of τ alone. To fix the parametrisation we impose an initial surface,

$$x(\sigma, \eta, 0) = \sigma, \qquad y(\sigma, \eta, 0) = \eta, \qquad t(\sigma, \eta, 0) = 0. \tag{2.78}$$

[1] I am grateful to Benjamin Sharp for the advice on applying the method of characteristics to three-variate problems that unlocked this section.

With G a function of τ only along a characteristic, the chain rule gives

$$\frac{\mathrm{d}G}{\mathrm{d}\tau} = \frac{\partial G}{\partial t}\frac{\mathrm{d}t}{\mathrm{d}\tau} + \frac{\partial G}{\partial x}\frac{\mathrm{d}x}{\mathrm{d}\tau} + \frac{\partial G}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}\tau},$$

which is to be compared with (2.75) written as

$$0 = -\frac{\partial G}{\partial t} + \mu(1-x)\frac{\partial G}{\partial x} + \alpha(x-y)\frac{\partial G}{\partial y}.$$

Comparing coefficients yields the characteristic system

$$\frac{\mathrm{d}G}{\mathrm{d}\tau} = 0, \tag{2.79}$$

$$\frac{\mathrm{d}t}{\mathrm{d}\tau} = -1, \tag{2.80}$$

$$\frac{\mathrm{d}x}{\mathrm{d}\tau} = \mu(1-x), \tag{2.81}$$

$$\frac{\mathrm{d}y}{\mathrm{d}\tau} = \alpha(x-y). \tag{2.82}$$

The equation for G, (2.79). G is constant along characteristics, so G is a function of (σ, η) alone. Under (2.78) the initial condition $G(x, y, 0) = y^N$ becomes $G(\sigma, \eta, 0) = \eta^N$, so

$$G(\sigma, \eta, \tau) = \eta^N. \tag{2.83}$$

The whole problem thus reduces to expressing η in terms of (x, y, t).

The equation for t, (2.80). Integrating gives $t = -\tau + c(\sigma, \eta)$, and (2.78) fixes $c = 0$:

$$t = -\tau. \tag{2.84}$$

The characteristic parameter runs backwards in time, which is what allows the initial surface to be reached from any (x, y, t).

The equation for x, (2.81). Separating variables gives $1 - x = c(\sigma, \eta)\mathrm{e}^{-\mu\tau}$, and (2.78) gives $c = 1 - \sigma$:

$$1 - x = (1-\sigma)\mathrm{e}^{-\mu\tau}. \tag{2.85}$$

The equation for y, (2.82). Substituting (2.85),

$$\frac{\mathrm{d}y}{\mathrm{d}\tau} = \alpha\big(1 - (1-\sigma)\mathrm{e}^{-\mu\tau} - y\big),$$

and solving with integrating factor $\mathrm{e}^{\alpha\tau}$,

$$y = 1 - \frac{\alpha}{\alpha-\mu}(1-\sigma)\mathrm{e}^{-\mu\tau} + c(\sigma, \eta)\mathrm{e}^{-\alpha\tau}.$$

Applying (2.78) gives $c(\sigma, \eta) = \eta + \frac{\alpha}{\alpha-\mu}(1-\sigma) - 1$, so

$$y = 1 - \frac{\alpha}{\alpha-\mu}(1-\sigma)\mathrm{e}^{-\mu\tau} + \left(\eta + \frac{\alpha}{\alpha-\mu}(1-\sigma) - 1\right)\mathrm{e}^{-\alpha\tau}. \tag{2.86}$$

Recovering the solution. Rearranging (2.86) for η, substituting σ from (2.85) and $\tau = -t$ from (2.84), gives

$$\eta = 1 - \frac{\mu}{\mu-\alpha}\mathrm{e}^{-\alpha t} + \frac{\alpha}{\mu-\alpha}\mathrm{e}^{-\mu t} + \frac{\alpha}{\mu-\alpha}(\mathrm{e}^{-\alpha t} - \mathrm{e}^{-\mu t})x + \mathrm{e}^{-\alpha t}y,$$

and (2.83) then reproduces (2.77) exactly. The two routes agree, and the machinery is ready for the case in which only one of them is available.

2.7.4 Absorption–birth–death

Introducing replication for particles inside the compartment, the independence property is still present but no longer serves as a shortcut: from a single initial particle, birth events make infinitely many states reachable, so $G_1(x, y, t)$ is an infinite sum and there is nothing to be gained by factorising. The partial differential equation must therefore be solved directly, and by now that can be done.

With X_t and Y_t as before, birth at per-capita rate λ is added to the transition probabilities:

$$\begin{aligned}
\mathbb{P}(\Delta X = 1,\ \Delta Y = -1 \mid Y_t = j) &= j\alpha\,\Delta t, && \text{(absorption)}\\
\mathbb{P}(\Delta X = 1,\ \Delta Y = 0 \mid X_t = i) &= i\lambda\,\Delta t, && \text{(birth)}\\
\mathbb{P}(\Delta X = -1,\ \Delta Y = 0 \mid X_t = i) &= i\mu\,\Delta t, && \text{(death)}\\
\mathbb{P}(\Delta X = 0,\ \Delta Y = 0 \mid X_t = i,\ Y_t = j) &= 1 - (j\alpha + i\mu + i\lambda)\Delta t, && \text{(no event)}
\end{aligned}$$

with forward Kolmogorov equation

$$\frac{\mathrm{d}p_{i,j}}{\mathrm{d}t} = \alpha(j+1)p_{i-1,j+1} + \mu(i+1)p_{i+1,j} + \lambda(i-1)p_{i-1,j} - (\alpha j + \mu i + \lambda i)p_{i,j}. \tag{2.87}$$

The same reasoning as before gives the generating-function equation

$$\frac{\partial G}{\partial t} = (\mu - (\mu+\lambda)x + \lambda x^2)\frac{\partial G}{\partial x} + \alpha(x-y)\frac{\partial G}{\partial y}, \qquad G(x, y, 0) = y^N, \tag{2.88}$$

whose x-coefficient is exactly the quadratic met in (2.13) for the linear birth–death process. Reparametrising $(x, y, t) \to (\sigma, \eta, \tau)$ as in section 2.7.3 gives the characteristic system

$$\frac{\mathrm{d}G}{\mathrm{d}\tau} = 0, \tag{2.89}$$

$$\frac{\mathrm{d}t}{\mathrm{d}\tau} = -1, \tag{2.90}$$

$$\frac{\mathrm{d}x}{\mathrm{d}\tau} = \mu - (\mu+\lambda)x + \lambda x^2, \tag{2.91}$$

$$\frac{\mathrm{d}y}{\mathrm{d}\tau} = \alpha(x-y), \tag{2.92}$$

with the same initial conditions (2.78).

The equations for G and t, (2.89)–(2.90). As before,

$$G(\sigma, \eta, \tau) = \eta^N, \qquad t = -\tau. \tag{2.93}$$

The equation for x, (2.91). This is a Riccati equation of the same shape as the one governing the survival probability in section 2.4.4. Its right-hand side factorises, and the roots are elementary rather than the surd expression that mechanical use of the quadratic formula suggests:

$$\mu - (\mu+\lambda)x + \lambda x^2 = \lambda(x-x_+)(x-x_-), \qquad x_+ = 1, \quad x_- = \frac{\mu}{\lambda}. \tag{2.94}$$

The discriminant is $\left(\frac{\mu+\lambda}{2\lambda}\right)^2 - \frac{\mu}{\lambda} = \left(\frac{\lambda-\mu}{2\lambda}\right)^2$, a perfect square, so the surds cancel. It is convenient to name the combination that appears throughout,

$$b_1 = \lambda(x_+ - x_-) = \lambda - \mu, \tag{2.95}$$

the net interior growth rate. Separating variables in (2.91) and applying (2.78) gives the characteristic in the compact form

$$\frac{x-x_+}{x-x_-} = \left(\frac{\sigma-x_+}{\sigma-x_-}\right) \mathrm{e}^{b_1\tau}, \tag{2.96}$$

or, solved for x,

$$x(\tau) = \frac{x_+(\sigma-x_-) - x_-(\sigma-x_+)\mathrm{e}^{b_1\tau}}{(\sigma-x_-) - (\sigma-x_+)\mathrm{e}^{b_1\tau}}. \tag{2.97}$$

The equation for y, (2.92). The last characteristic equation is $\mathrm{d}y/\mathrm{d}\tau = \alpha(x-y)$, which with an integrating factor becomes

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\left(\mathrm{e}^{\alpha\tau}y\right) = \alpha\,\mathrm{e}^{\alpha\tau}x(\tau). \tag{2.98}$$

The integral on the right is where the difficulty lies. Rather than substitute (2.97) and split the resulting quotient — which leads to two hypergeometric integrals and a good deal of bookkeeping — it is cleaner to expand $x(\tau)$ first. Writing $x - x_- = (x-x_+) + (x_+ - x_-)$ in (2.96) and solving for x gives

$$x(\tau) = x_+ + (x_+ - x_-)\frac{\zeta(\tau)}{1-\zeta(\tau)}, \qquad \zeta(\tau) = \left(\frac{\sigma-x_+}{\sigma-x_-}\right)\mathrm{e}^{b_1\tau}, \tag{2.99}$$

so that, for $|\zeta| < 1$,

$$x(\tau) = x_+ + (x_+ - x_-)\sum_{n=1}^{\infty}\zeta(\tau)^n.$$

Every term is now a pure exponential in τ and integrates immediately. Writing $Z = (\sigma-x_+)/(\sigma-x_-)$, so that $\zeta(\tau) = Z\mathrm{e}^{b_1\tau}$ and $Z^n\mathrm{e}^{nb_1\tau} = \zeta^n$,

$$\begin{aligned}
\alpha\int \mathrm{e}^{\alpha\tau}x(\tau)\,\mathrm{d}\tau &= \alpha x_+\int \mathrm{e}^{\alpha\tau}\mathrm{d}\tau + \alpha(x_+ - x_-)\sum_{n=1}^{\infty} Z^n \int \mathrm{e}^{(\alpha+nb_1)\tau}\mathrm{d}\tau \\
&= \mathrm{e}^{\alpha\tau}\left[x_+ + \alpha(x_+ - x_-)\sum_{n=1}^{\infty}\frac{\zeta^n}{\alpha+nb_1}\right] + \text{const.}
\end{aligned}$$

Setting $\delta = \alpha/b_1$ and using $\alpha(x_+ - x_-)/b_1 = \alpha/\lambda$ by (2.95), this defines the function that carries the whole solution:

$$\Psi(\zeta) = x_+ + \frac{\alpha}{\lambda}\sum_{n=1}^{\infty}\frac{\zeta^n}{n+\delta}, \qquad \delta = \frac{\alpha}{b_1} = \frac{\alpha}{\lambda-\mu}. \tag{2.100}$$

So $\alpha\int \mathrm{e}^{\alpha\tau}x(\tau)\mathrm{d}\tau = \mathrm{e}^{\alpha\tau}\Psi(\zeta(\tau)) + \text{const}$, and integrating (2.98) gives $\mathrm{e}^{\alpha\tau}y = \mathrm{e}^{\alpha\tau}\Psi(\zeta(\tau)) + \text{const}$. The initial condition $y(\sigma,\eta,0) = \eta$, at which $\zeta(0) = Z$, fixes the constant as $\eta - \Psi(Z)$, so

$$\eta = \mathrm{e}^{\alpha\tau}\Big[y - \Psi\big(\zeta(\tau)\big)\Big] + \Psi(Z). \tag{2.101}$$

Finally, put $\tau = -t$ and use (2.96): writing

$$W = \frac{x - x_+}{x - x_-}$$

we have $\zeta(\tau) = W$ and $Z = W\mathrm{e}^{-b_1\tau} = W\mathrm{e}^{b_1 t}$. Substituting into (2.101) and applying (2.93) gives the solution.

Proposition 2.13 (Generating function for absorption–birth–death)**.** *With $x_+ = 1$, $x_- = \mu/\lambda$, $b_1 = \lambda - \mu$, $\delta = \alpha/b_1$, $W = (x - x_+)/(x - x_-)$ and Ψ as in (2.100),*

$$\boxed{G(x, y, t) = \Big\{\mathrm{e}^{-\alpha t}\big[y - \Psi(W)\big] + \Psi\big(W\mathrm{e}^{b_1 t}\big)\Big\}^N.} \tag{2.102}$$

Three checks confirm the result. At $t = 0$ the two Ψ terms cancel and $G = y^N$, the required initial condition. At $x = y = 1$ we have $W = 0$ and $\Psi(0) = x_+ = 1$, so $G = 1$ and probability is conserved. And setting $x = 1$ alone gives

$$G(1, y, t) = \big(\mathrm{e}^{-\alpha t}y + 1 - \mathrm{e}^{-\alpha t}\big)^N,$$

so $Y_t \sim \mathrm{Binomial}(N, \mathrm{e}^{-\alpha t})$ — exactly right, since the exterior particles are absorbed independently at rate α and what happens inside the compartment cannot affect them. Beyond these, (2.102) has been verified numerically against direct integration of the master equation (2.87), agreeing to around 10^{-13} across sub- and supercritical interior regimes and for $N = 1, 2, 3$.

Hypergeometric form

The series in (2.100) is a Gauss hypergeometric function in disguise. With

$${}_2F_1(a, b; c; z) = \sum_{n=0}^{\infty} \frac{(a)_n (b)_n}{(c)_n} \frac{z^n}{n!}, \qquad (q)_n = \begin{cases} 1, & n = 0, \\ q(q+1)\cdots(q+n-1), & n \geq 1, \end{cases} \tag{2.103}$$

where $(q)_n$ is the rising Pochhammer symbol [18], the case $a = 1$, $b = \delta$, $c = \delta + 1$ collapses to a single ratio. Since $(1)_n = n!$ and

$$\frac{(\delta)_n}{(\delta+1)_n} = \frac{\delta(\delta+1)\cdots(\delta+n-1)}{(\delta+1)\cdots(\delta+n-1)(\delta+n)} = \frac{\delta}{\delta+n}, \tag{2.104}$$

we have

$${}_2F_1(1, \delta; \delta+1; z) = \sum_{n=0}^{\infty} \frac{\delta}{\delta+n} z^n. \tag{2.105}$$

Comparing with (2.100) and using $\alpha/(\lambda\delta) = b_1/\lambda = x_+ - x_-$,

$$\Psi(\zeta) = x_+ + (x_+ - x_-)\Big[{}_2F_1(1, \delta; \delta+1; \zeta) - 1\Big]. \tag{2.106}$$

(2.102) together with (2.106) is the closed form. It is not pretty, but it is a genuine closed form in terms of a standard special function, and the elementary series (2.100) is what one would actually evaluate. An alternative derivation of the same object, by way of the integral $\int \mathrm{e}^{h\tau}/(P + Q\mathrm{e}^{k\tau})\,\mathrm{d}\tau$, is given in section 2.A; that route is longer, but it produces a general identity of independent use.

Remark 2.14 (Domain of validity). The series (2.100) converges for $|\zeta| < 1$, and the arguments appearing in (2.102) are W and $We^{b_1 t}$. In the subcritical interior regime $\mu > \lambda$ we have $b_1 < 0$ and $x_- = \mu/\lambda > 1$, so for $x \in [0, 1]$ the argument W lies in $[0, \lambda/\mu] \subset [0, 1)$ and $We^{b_1 t}$ is smaller still: the series representation is valid on the whole of the unit square, which is the case of most interest here. When $\lambda > \mu$ the series converges only for x above the smaller fixed point μ/λ, and elsewhere one must use the analytic continuation of ${}_2F_1$ — conveniently, the Euler integral

$$
{}_2F_1(1, \delta; \delta + 1; \zeta) = \delta \int_0^1 \frac{u^{\delta-1}}{1 - \zeta u} \, \mathrm{d}u \qquad (\delta > 0,\ \zeta \notin [1, \infty)),
$$

which was used for the numerical verification quoted above.

Remark 2.15 (Resonance). The denominators in (2.100) vanish when $n + \delta = 0$ for some integer $n \geq 1$, that is when

$$
\alpha = n(\mu - \lambda), \qquad n = 1, 2, \ldots
$$

This is the classical resonance condition for linearising the y-equation: the decay rate α of the absorption channel coincides with a harmonic of the interior relaxation rate $\mu - \lambda$. The generating function itself is perfectly regular there, being analytic in all parameters, so the singularity is removable, and at resonance the corresponding term acquires a factor τ — a secular term — in place of the simple exponential. The resonant parameter values form a set of measure zero and the limit may be taken numerically, but the degeneracy should be kept in mind when α happens to sit near an integer multiple of $\mu - \lambda$.

2.8 Summary

Three things have been assembled in this chapter, and it is worth separating them before they are used.

The first is standard apparatus: exponential clocks and the races between them, continuous-time Markov chains and their generators, generating functions, and the linear birth–death process together with the first-order equation (2.13) satisfied by its generating function. That equation is the template for every generating-function calculation in this thesis.

The second is the quasi-stationary picture. A subcritical binary Galton–Watson process is certain to die out, but conditioned on survival its population settles to a fixed distribution whose mean is $1/A(p)$ and whose variance is (2.34). Both are expressed entirely in terms of p and the constant $A(p) = \lim_n S_n/(2p)^n$. That constant admits the exact product (2.52) and the exact series (2.53); it is confined by proposition 2.7 between half the continuous-time constant $A_c(p)$ and $2\varepsilon/(1 + 2\varepsilon)$, approaching the first as $p \to 0$ and the second as $p \to \frac{1}{2}^-$, where $A(p) \sim 2(1 - 2p)$; and it is a value of the Koenigs linearising function of the logistic map, which is why the search for an elementary formula failed and should be expected to. (2.68) is the scheme used to evaluate it in the rest of this thesis.

The third is the method of characteristics for the generating-function equations of absorption models, developed on a case whose answer was already known and then applied to absorption–birth–death, where it produces the closed form (2.102). That is the tool that the later chapters on birth–death–catastrophe processes take up, first for the process of definition 2.5 and then for the multi-type and multi-compartment extensions.

What has deliberately not been done here is to develop those extensions. The birth–death–catastrophe process has been defined and its generating-function equation written down, and no more; the results belong to the chapters that need them.

2.A An integral identity for the hypergeometric function

The route to (2.100) taken in section 2.7.4 expands the characteristic $x(\tau)$ in a geometric series and integrates term by term. An alternative is to integrate the quotient (2.97) directly, splitting the numerator into two pieces, each of the form

$$I(\tau) = \int \frac{\mathrm{e}^{h\tau}}{P + Q\,\mathrm{e}^{k\tau}}\,\mathrm{d}\tau \tag{2.107}$$

with h, k, P, Q constant. That route is longer and easier to get wrong — the two pieces carry different prefactors, and dropping one produces a formula that still satisfies the initial condition while violating $G(1,1,t) = 1$ — but the identity it rests on is useful in its own right, so it is recorded and proved here.

Proposition 2.16. *For $h/k > 0$ and $-\frac{Q}{P}\mathrm{e}^{k\tau} \notin [1,\infty)$,*

$$\int \frac{\mathrm{e}^{h\tau}}{P + Q\mathrm{e}^{k\tau}}\,\mathrm{d}\tau = \frac{\mathrm{e}^{h\tau}}{Ph}\,{}_2F_1\left(1, \frac{h}{k}; \frac{h}{k} + 1; -\frac{Q}{P}\mathrm{e}^{k\tau}\right) + \textit{const.} \tag{2.108}$$

Proof. We use the Euler integral representation of the hypergeometric function [18],

$${}_2F_1(a,b;c;z) = \frac{\Gamma(c)}{\Gamma(b)\Gamma(c-b)} \int_0^1 u^{b-1}(1-u)^{c-b-1}(1-zu)^{-a}\,\mathrm{d}u, \tag{2.109}$$

Γ being the gamma function.

Start from (2.107) and substitute $v = \mathrm{e}^{k\tau}$, so that $\mathrm{d}\tau = k^{-1}v^{-1}\mathrm{d}v$:

$$I = \frac{1}{k}\int \frac{v^{h/k}}{P + Qv}\,v^{-1}\mathrm{d}v = \frac{1}{Pk}\int \frac{v^{h/k-1}}{1 + \frac{Q}{P}v}\,\mathrm{d}v.$$

Fix the antiderivative by writing this as a definite integral from 0 to v in a dummy variable w; the integral converges at the lower limit precisely because $h/k > 0$:

$$I = \frac{1}{Pk}\int_0^v \frac{w^{h/k-1}}{1 + \frac{Q}{P}w}\,\mathrm{d}w.$$

Now rescale, $w = vu$, so that $\mathrm{d}w = v\,\mathrm{d}u$ with v held fixed:

$$I = \frac{1}{Pk}\int_0^1 \frac{(vu)^{h/k-1}}{1 + \frac{Q}{P}vu}\,v\,\mathrm{d}u = \frac{v^{h/k}}{Pk}\int_0^1 \frac{u^{h/k-1}}{1 + \frac{Q}{P}vu}\,\mathrm{d}u. \tag{2.110}$$

The remaining integrand matches that of (2.109) with $b = h/k$, $c = h/k + 1$, $a = 1$ and $z = -\frac{Q}{P}v$, since then $(1-u)^{c-b-1} = 1$ and $(1-zu)^{-a} = \left(1 + \frac{Q}{P}vu\right)^{-1}$. Hence, using $\Gamma(z+1) = z\Gamma(z)$,

$$\int_0^1 \frac{u^{h/k-1}}{1 + \frac{Q}{P}vu}\,\mathrm{d}u = \frac{\Gamma\left(\frac{h}{k}\right)\Gamma(1)}{\Gamma\left(\frac{h}{k}+1\right)}\,{}_2F_1\left(1, \frac{h}{k}; \frac{h}{k} + 1; -\frac{Q}{P}v\right) = \frac{k}{h}\,{}_2F_1\left(1, \frac{h}{k}; \frac{h}{k} + 1; -\frac{Q}{P}v\right).$$

Substituting into (2.110) gives

$$I = \frac{v^{h/k}}{Ph}\,{}_2F_1\left(1, \frac{h}{k}; \frac{h}{k} + 1; -\frac{Q}{P}v\right),$$

and restoring $v = \mathrm{e}^{k\tau}$ gives (2.108). $\square$

Remark 2.17. Applied to the two pieces of (2.97) — the first with $h = \alpha$, the second with $h = \alpha + b_1$, and both with $k = b_1$, $P = \sigma - x_-$ and $Q = -(\sigma - x_+)$ — proposition 2.16 reproduces (2.106) once the prefactors $\alpha x_+(\sigma - x_-)$ and $-\alpha x_-(\sigma - x_+)$ are carried through correctly. The factor $(\sigma - x_+)/(\sigma - x_-)$ attached to the second piece is easily lost, and its loss is invisible at $t = 0$; checking $G(1,1,t) = 1$ catches it immediately.

2.B Extracting state probabilities from a generating function

A generating function defined by a formula rather than as an explicit series — for example

$$G(x,y,t) = \Big((1 - \mathrm{e}^{-\alpha t})x + \mathrm{e}^{-\alpha t} y\Big)^n \tag{2.111}$$

— can always be converted back to the form

$$G(x,y,t) = \sum_{i=0}^{\infty}\sum_{j=0}^{\infty} p_{i,j}(t)\, x^i y^j, \tag{2.112}$$

recovering the time-dependent state probabilities. All that is required is that G be repeatedly partially differentiable in x and y.

The reason is the bivariate Maclaurin expansion,

$$f(x,y) = \sum_{k=0}^{\infty} \frac{1}{k!} \sum_{\varrho=0}^{k} \binom{k}{\varrho} \partial_x^{\varrho} \partial_y^{k-\varrho} f\Big|_{(0,0)}\; x^{\varrho} y^{k-\varrho},$$

whose coefficient of $x^i y^j$, taking $k = i + j$ and $\varrho = i$, is $\frac{1}{(i+j)!}\binom{i+j}{i}\partial_x^i \partial_y^j f\big|_{(0,0)}$. Hence

$$p_{i,j}(t) = \frac{1}{i!\, j!}\, \partial_x^i \partial_y^j G\Big|_{(0,0)}. \tag{2.113}$$

The multilinear case

Both (2.73) and (2.77) have the form

$$G(x,y,t) = \big(c(t) + f(t)\, x + g(t)\, y\big)^n, \tag{2.114}$$

for which the derivatives are immediate. Each differentiation lowers the power by one and brings down a factor, so

$$\partial_x^a \partial_y^b G = \frac{n!}{(n-a-b)!}\, f^a g^b \big(c + fx + gy\big)^{n-a-b} \qquad (a + b \le n), \tag{2.115}$$

and zero for $a + b > n$. Evaluating at the origin and applying (2.113),

$$p_{i,j}(t) = \frac{n!}{i!\, j!\, (n-i-j)!}\; f(t)^i g(t)^j c(t)^{\,n-i-j}, \qquad i + j \le n, \tag{2.116}$$

and $p_{i,j} = 0$ otherwise. This is the trinomial distribution, which is exactly what it should be: the n particles are independent, and at time t each is in one of three states, with probabilities c (neither inside nor outside — that is, dead), f (inside) and g (outside). Note in particular that the falling factorial $n!/(n-i-j)!$ in (2.115) is *not* n^{i+j}; the two agree only to leading order in n, and using the latter breaks normalisation.

Applied to the absorption models

For absorption only, (2.111), we have $c = 0$, $f = 1 - \mathrm{e}^{-\alpha t}$ and $g = \mathrm{e}^{-\alpha t}$. The factor c^{n-i-j} then vanishes unless $i + j = n$, and (2.116) gives

$$p_{i,j}(t) = \binom{n}{i} (1 - \mathrm{e}^{-\alpha t})^{i} (\mathrm{e}^{-\alpha t})^{j}, \qquad i + j = n,$$

so $X_t \sim \mathrm{Binomial}(n, 1 - \mathrm{e}^{-\alpha t})$ and $Y_t = n - X_t$, as anticipated in section 2.7.1.

For absorption–death, (2.77), the three functions are

$$c(t) = 1 - \frac{\mu}{\mu - \alpha}\mathrm{e}^{-\alpha t} + \frac{\alpha}{\mu - \alpha}\mathrm{e}^{-\mu t}, \qquad f(t) = \frac{\alpha}{\mu - \alpha}\big(\mathrm{e}^{-\alpha t} - \mathrm{e}^{-\mu t}\big), \qquad g(t) = \mathrm{e}^{-\alpha t},$$

which are precisely $p_{0,0}$, $p_{1,0}$ and $p_{0,1}$ of (2.76), so that

$$p_{i,j}(t) = \frac{n!}{i!\, j!\, (n - i - j)!}\, f(t)^{i} g(t)^{j} c(t)^{\,n-i-j}.$$

Each particle independently ends up dead, absorbed, or still outside, with those three probabilities, and the joint law is trinomial. That the generating function factorises in this way is (2.72) read backwards.

The absorption–birth–death generating function of proposition 2.13 is not of the form (2.114), and there (2.113) must be applied to (2.102) directly, or the coefficients extracted numerically by Cauchy integrals on circles $|x| = |y| = \varrho < 1$.

Bibliography

[1] James R. Norris. *Markov Chains*. Cambridge Series in Statistical and Probabilistic Mathematics. Cambridge University Press, Cambridge, 1997.

[2] Samuel Karlin and Howard M. Taylor. *A First Course in Stochastic Processes*. Academic Press, New York, 2nd edition, 1975.

[3] Linda J. S. Allen. *An Introduction to Stochastic Processes with Applications to Biology*. CRC Press, Boca Raton, FL, 2nd edition, 2010.

[4] Daniel T. Gillespie. Exact stochastic simulation of coupled chemical reactions. *The Journal of Physical Chemistry*, 81(25):2340–2361, 1977.

[5] David G. Kendall. On the generalized "birth-and-death" process. *The Annals of Mathematical Statistics*, 19(1):1–15, 1948.

[6] Theodore E. Harris. *The Theory of Branching Processes*. Springer, Berlin, 1963.

[7] Krishna B. Athreya and Peter E. Ney. *Branching Processes*. Springer, Berlin, 1972.

[8] Akiva M. Yaglom. Certain limit theorems of the theory of branching random processes. *Doklady Akademii Nauk SSSR (N.S.)*, 56:795–798, 1947.

[9] Sylvie Méléard and Denis Villemonais. Quasi-stationary distributions and population processes. *Probability Surveys*, 9:340–410, 2012.

[10] Pierre Collet, Servet Martínez, and Jaime San Martín. *Quasi-Stationary Distributions: Markov Chains, Diffusions and Dynamical Systems*. Springer, Berlin, 2013.

[11] Graham E. Budd and Richard P. Mann. History is written by the victors: The effect of the push of the past on the fossil record. *Evolution*, 72(11):2276–2291, 2018.

[12] Kenneth O Stanley and Risto Miikkulainen. Evolving neural networks through augmenting topologies. *Evolutionary Computation*, 10(2):99–127, 2002.

[13] Benjamin G Cohen, Burcu Beykal, and George M Bollas. Physics-informed genetic programming for discovery of partial differential equations from scarce and noisy data. *Journal of Computational Physics*, 514:113261, 2024.

[14] Gabriel Koenigs. Recherches sur les intégrales de certaines équations fonctionnelles. *Annales Scientifiques de l'École Normale Supérieure*, 1:3–41, 1884.

[15] John Milnor. *Dynamics in One Complex Variable*. Princeton University Press, Princeton, NJ, 3rd edition, 2006.

[16] Lennart Carleson and Theodore W. Gamelin. *Complex Dynamics*. Springer, New York, 1993.

[17] Paul-Georg Becker and Walter Bergweiler. Hypertranscendency of conjugacies in complex dynamics. *Mathematische Annalen*, 301(1):463–468, 1995.

[18] Frank W. J. Olver, Daniel W. Lozier, Ronald F. Boisvert, and Charles W. Clark, editors. *NIST Handbook of Mathematical Functions*. Cambridge University Press, Cambridge, 2010.